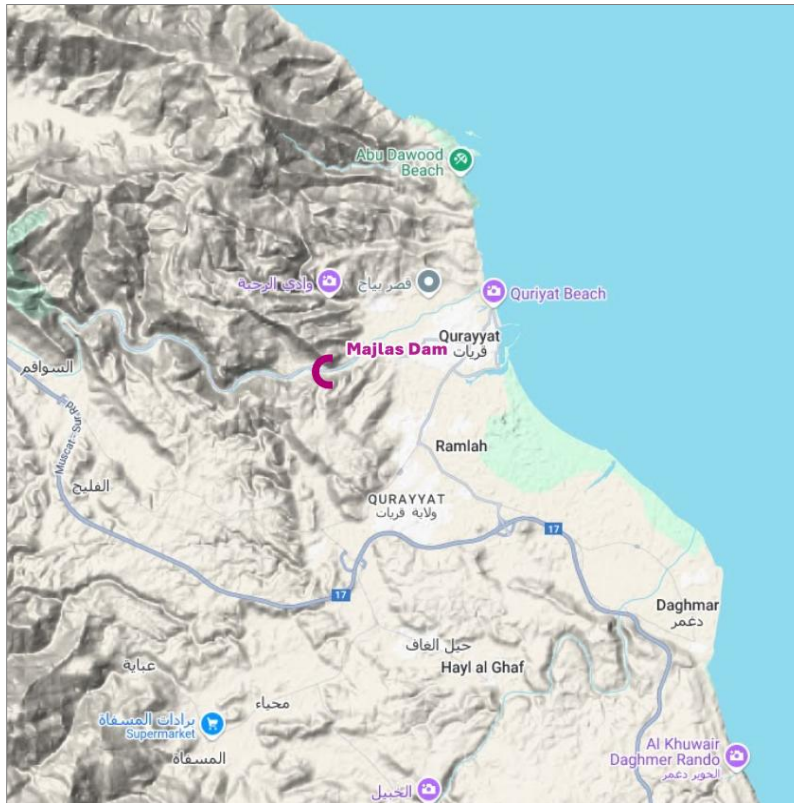




Sultanate of Oman

Ministry of Agriculture, Fisheries Wealth & Water Resources

Consultancy Services for Design Review and Supervision of Wadi Majlas Dam in Wilayat Qurayat, Muscat Governorate



Wadi Majlas Flood Protection Dam Dam Break Analysis Report

Rev 00

September 2026



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ABBREVIATIONS AND NOMENCLATURE

Terms and short names used in this report.

Table 1 - Abbreviations and terms.

Term	Meaning
AIDR	Australian Institute for Disaster Resilience; its flood hazard classes H1 to H6 are used here
Breach	The opening that forms in the dam when it fails
DTM	Digital terrain model, the ground surface used by the model
EAP	Emergency action plan
FSL	Full supply level, 84.5 m a.s.l., the spillway crest
HEC-RAS	Hydrologic Engineering Center River Analysis System, version 6.6, the hydraulic model
Monolith	One concrete block of the dam between two vertical joints; the dam has 16 of them
PAR	Population at risk: people living in the flooded area
PMF, PMP	Probable maximum flood, probable maximum precipitation
Restart file	A saved state of the model (water levels everywhere) used as the starting point of a run
S1, S2, S3	Sunny-day failure; flood-day failure during the PMF; PMF without failure
D, N	With the downstream training dikes; without them
m a.s.l.	Metres above sea level



EXECUTIVE SUMMARY

This report presents the dam break analysis of the Wadi Majlas Flood Protection Dam, a 72 m high roller-compacted concrete gravity dam about 2 km upstream of Qurayat. It applies the methodology of the Dam Break Analysis Methodology Report Rev 00 (September 2026) with the HEC-RAS 6.6 two-dimensional model built for this study. Three scenarios were run with the downstream training dikes in place, the no-dikes condition was checked on the sunny-day scenario, and eight sensitivity runs bound the breach parameters and the flood.

Sunny-day failure (S1D). With the reservoir at the full supply level and no flood, the sudden failure of five monoliths (85 m of the dam) releases 103 Mm^3 in about two hours. The peak outflow is $45,312 \text{ m}^3/\text{s}$, reached 10 min after the failure begins. Water deeper than 0.3 m covers 17.0 km^2 of the coastal plain; the plain is reached within 15 min, the shoreline within 22 min and the last corner of the flooded area after 1 h 25 min. About 23,212 people live in the flooded area, 22,541 of them where the flood is unsafe for people and vehicles or worse; 15,579 of them are reached within 15 minutes of the breach and 21,739 within 30.

PMF without failure (S3D). The reservoir peaks at 96.06 m a.s.l. and the spillway passes $17,323 \text{ m}^3/\text{s}$. The dam is not overtopped.

Flood-day failure (S2D). The dam is assumed to fail when the reservoir reaches its maximum level during the PMF, 15 h 15 min into the storm. The peak outflow rises from $17,323 \text{ m}^3/\text{s}$ without failure to $63,858 \text{ m}^3/\text{s}$ with failure, and the flooded area from 25.9 to 27.1 km^2 . The failure wave adds 0.3 m or more to the water already standing on 25,365 people within 15 minutes of the breach.

Effect of the training dikes. The dikes were designed for the 200-year wadi flood, not for a dam break. In the sunny-day failure the flooded area is 17.0 km^2 with the dikes and 17.1 km^2 without; the population at risk is 23,212 and 23,299 respectively.

The results are meant for the emergency action plan and for the design review: the arrival times and hazard maps show where warning and evacuation are needed first, and the no-failure runs confirm the spillway capacity against the PMF. Figure 1 puts the three base scenarios side by side.



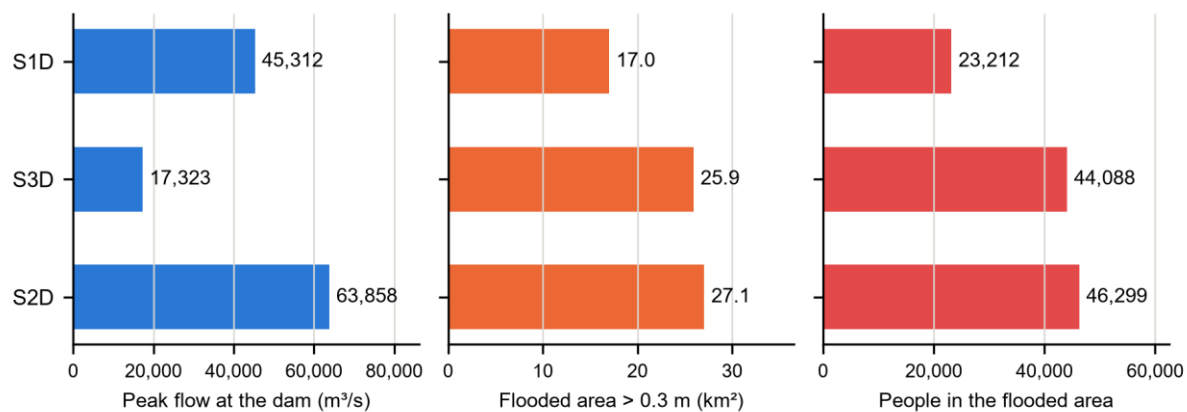


Figure 1 - The base scenarios at a glance: peak flow at the dam, flooded area and people in it.



1. INTRODUCTION

1.1. PURPOSE AND SCOPE

A dam break analysis answers three questions: how fast and how far would the water go if the dam failed, who and what would be in its way, and how much worse is a failure than the flood the dam is already passing. This report answers them for the Wadi Majlas dam with a two-dimensional hydraulic model, and gives the numbers that the emergency action plan¹ needs.

The methodology was set out in the Dam Break Analysis Methodology Report Rev 00 (September 2026). This report applies it. Chapter 2 condenses the method; where the application had to deviate from Rev 00, the deviation and its reason are stated in chapter 4.

1.2. THE DAM AND THE VALLEY IN BRIEF

The dam is a roller-compacted concrete (RCC) gravity dam with a grout-enriched RCC facing, 72 m above the wadi bed, 312 m long at the crest (95.5 m a.s.l.), with a 1.2 m parapet to 96.7 m a.s.l. The ungated ogee spillway, 200 m net length at 84.5 m a.s.l., is designed for 17,000 m³/s. The reservoir holds 102.2 Mm³ at the full supply level. Downstream, the wadi leaves a gorge and crosses the coastal plain of Qurayat, where two training dikes of 2.6 km guide the 200-year flood to the sea.

¹ Emergency action plan (EAP): the document that tells the operator and the authorities what to do, and whom to warn, when the dam is in danger. The arrival times and hazard maps in this report are its main technical input.



The HEC-RAS model: 2D area, boundary lines and dam connection

Wadi Majlas Flood Protection Dam, Dam Break Analysis | Renardet S.A. & Partners, 2026

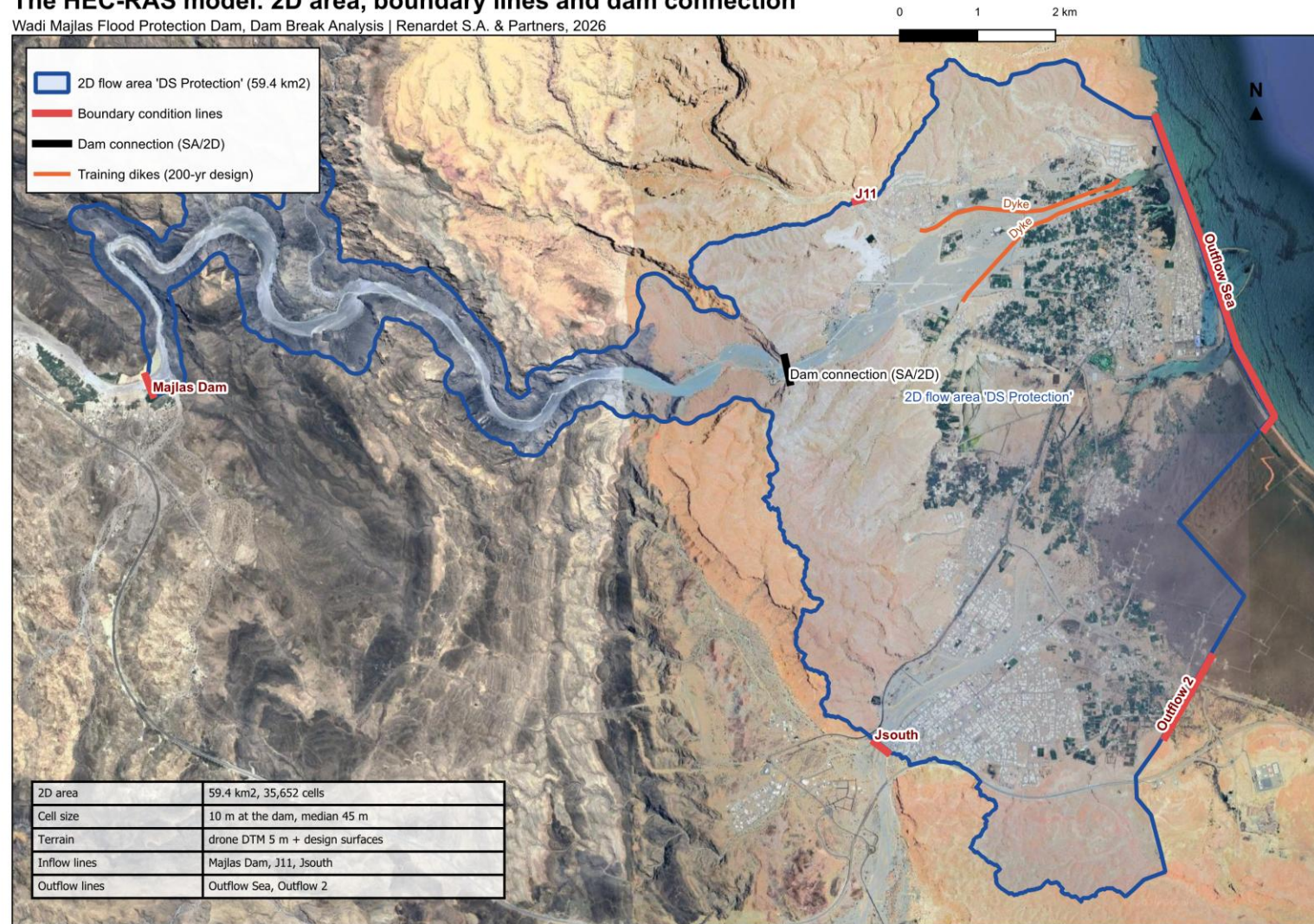


Figure 2 - The model: 2D flow area, boundary condition lines, dam connection and training dikes.

2. METHODOLOGY AS APPLIED

This chapter condenses the methodology report to what was actually done, so that this document can be read alone. The full reasoning, the review of guidance documents and the comparison of dam types are in Rev 00.

2.1. FAILURE MODE OF A CONCRETE GRAVITY DAM

A roller-compacted concrete gravity dam does not erode. It fails, if at all, by the sudden sliding or overturning of one or more monoliths, the blocks between vertical contraction joints, when the water load exceeds what the foundation contact can hold. The breach is therefore rectangular, vertical-sided, essentially instantaneous, and its width is a whole number of monoliths. The regression equations used for earth and rockfill dams, which predict an eroding trapezoidal breach over hours, do not apply; the FERC and USACE guidance for concrete gravity dams is followed instead: breach width up to half the crest length, formation time of a few minutes, and a sensitivity band on both.



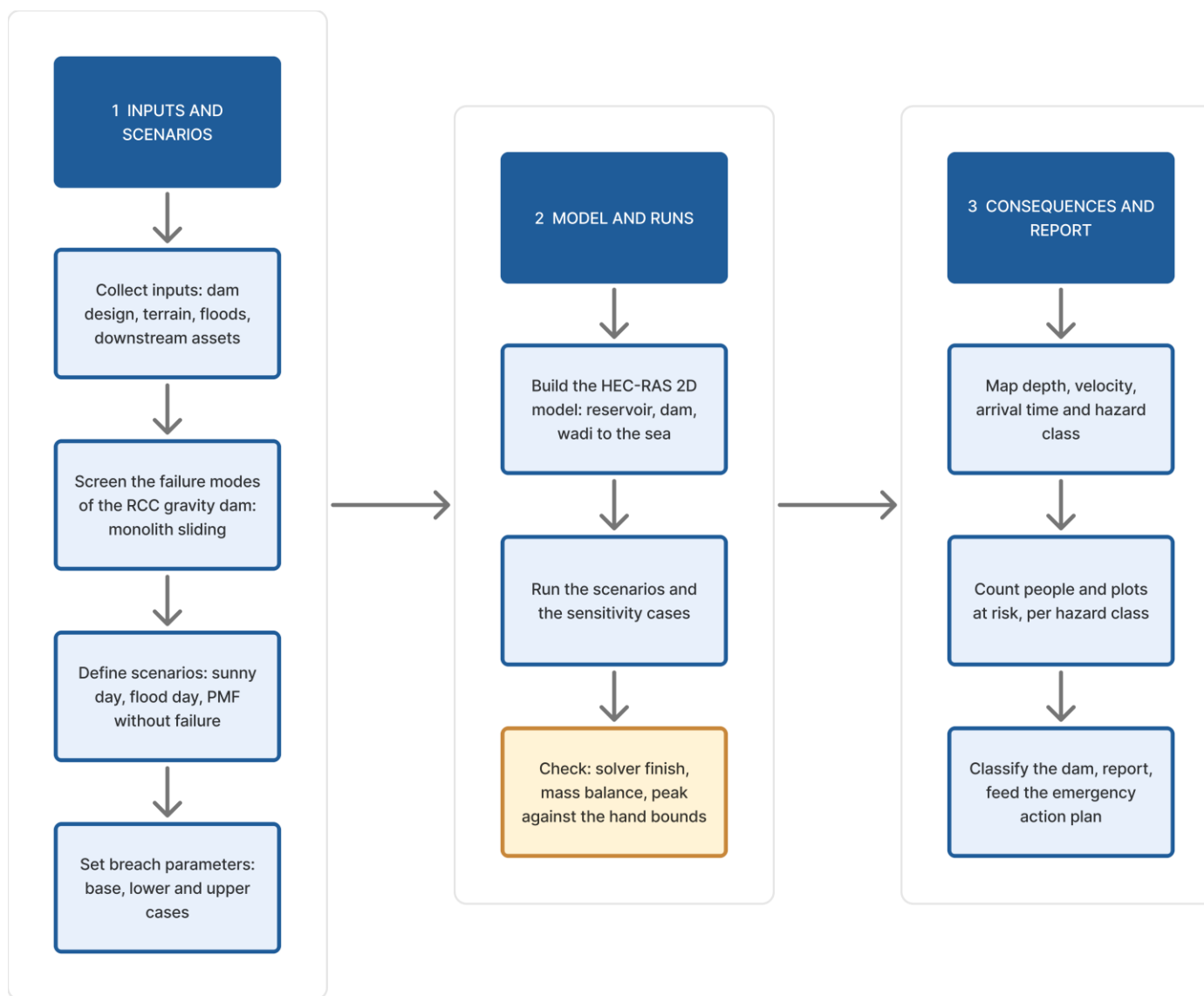


Figure 3 - The dam break analysis process, from data to emergency-plan inputs.



Two hand calculations bound the peak outflow before any model is trusted. The weir equation applied to the fully open breach at the starting pool gives the upper value:

$$Q_p = CB(H - Z_b)^{1.5} \quad (1)$$

where:

Q_p peak outflow through the breach (m^3/s)

C weir coefficient of the opening, $1.66 \text{ m}^{0.5}/\text{s}$ in the base case

B breach width, 85 m in the base case

H pool level at failure (m a.s.l.)

Z_b breach bottom, 23.0 m a.s.l.

The Ritter solution for the sudden removal of a wall onto a dry bed gives the lower value, because it accounts for the drop of the water surface at the opening:

$$Q_p = \frac{8}{27} B \sqrt{g} (H - Z_b)^{1.5} \quad (2)$$

where:

g gravity, 9.81 m/s^2

For the base breach and the full supply level the two give 71,000 and 38,000 m^3/s ; the modelled sunny-day peak of 45,312 m^3/s lies between them, lower than the weir value because the pool drops during the six minutes of formation and the tailwater rises in the stilling basin.

2.2. SCENARIOS AND THE REFERENCE CASE

Two failure scenarios bracket the question. The sunny-day failure, with the reservoir at the full supply level and no flood, gives the sharpest wave and the least warning. The flood-day failure assumes the dam fails at the peak of the probable maximum flood, when the load is highest; the failure moment is read from the no-failure run of the same flood. The no-failure run is also the reference: only what the failure adds to it counts against the dam. The scenarios and their default conditions are summarised in Figure 4.



For every consequence measure, the dam's contribution is the difference between the failure run and its no-failure twin:

$$\Delta C = C_{failure} - C_{no\ failure} \quad (3)$$

where:

ΔC the dam's contribution to any consequence measure C: depth, area, people, hazard class

$C_{failure}, C_{no\ failure}$ the measure in the failure run and in its no-failure twin

2.3. HYDRAULIC MODEL

The flood is routed with HEC-RAS 6.6 in two dimensions over one mesh from the reservoir to the sea, with the dam as a weir connection whose profile carries the spillway and non-overflow crests and into which the breach is cut. The inflows and the rain come from the HEC-HMS model of the hydrology study; chapter 4 gives every input as built, and Figure 5 the order in which the model was assembled. The spillway follows the ogee equation and the solver keeps its time step below the Courant limit:

$$Q = C_s L H^{1.5} \quad (4)$$

where:

Q spillway discharge (m^3/s)

C_s discharge coefficient of the ogee, 2.2 in the connection

L net crest length, 200 m

H head above the crest at 84.5 m (m)



$$Cr = \frac{V\Delta t}{\Delta x} \quad (5)$$

where:

Cr Courant number, kept between 0.4 and 0.9 by halving and doubling the step

V flow velocity plus wave celerity (m/s)

Δt time step (s), 0.335 to 42.9 s

Δx cell size in the direction of flow (m)

2.4. CONSEQUENCES

For each run the model gives, on a 2 m grid, the maximum depth and velocity, their product, the time at which 0.3 m of water first arrives and the duration above that depth. Depth and velocity are combined into the six Australian (AIDR) flood hazard classes, from H1, safe for people and vehicles, to H6, where buildings fail; a cell takes the highest class that any of the three limits gives:

$$D \times V = \text{depth} \times \text{velocity} \quad (6)$$

where:

D maximum depth (m)

V maximum velocity (m/s)

D × V their product (m²/s), the measure of the force on a person or a wall

Table 2 - AIDR flood hazard classes: a cell is in the class whose limits it exceeds.

Class	D × V limit (m ² /s)	Depth limit (m)	Velocity limit (m/s)	Meaning
H1	0.3	0.3	2.0	Generally safe for people, vehicles and buildings
H2	0.6	0.5	2.0	Unsafe for small vehicles
H3	0.6	1.2	2.0	Unsafe for vehicles, children and the elderly
H4	1.0	2.0	2.0	Unsafe for people and vehicles
H5	4.0	4.0	4.0	Unsafe for people and vehicles; buildings vulnerable to structural damage
H6	above H5			Unsafe for everything; buildings fail



People are counted on the GHS-POP 2025 population grid and properties on the plot layer of the flood-risk study, within the flooded area and per hazard class:

$$PAR = \sum P_i \text{ for every cell } i \text{ with } D_i > 0.3 \text{ m} \quad (7)$$

where:

PAR population at risk

P_i people in population cell i (100 m)

D_i maximum depth at the centre of cell i (m)

2.5. SENSITIVITY AND CHECKS

The breach width, the formation time, the weir coefficient of the opening and the flood itself are varied one at a time from the base case, so that each result carries a bound. Whether the no-dikes geometry needed its own flood-day runs was decided by the sunny-day pair:

$$\frac{|X_N - X_D|}{X_D} < 5\% \quad (8)$$

where:

X flooded area, people in the flooded area and median arrival time

N, D without and with the training dikes

Every run was checked for a clean solver finish and mass conservation before its products were used. Figure 6 shows the sequence of runs as executed.



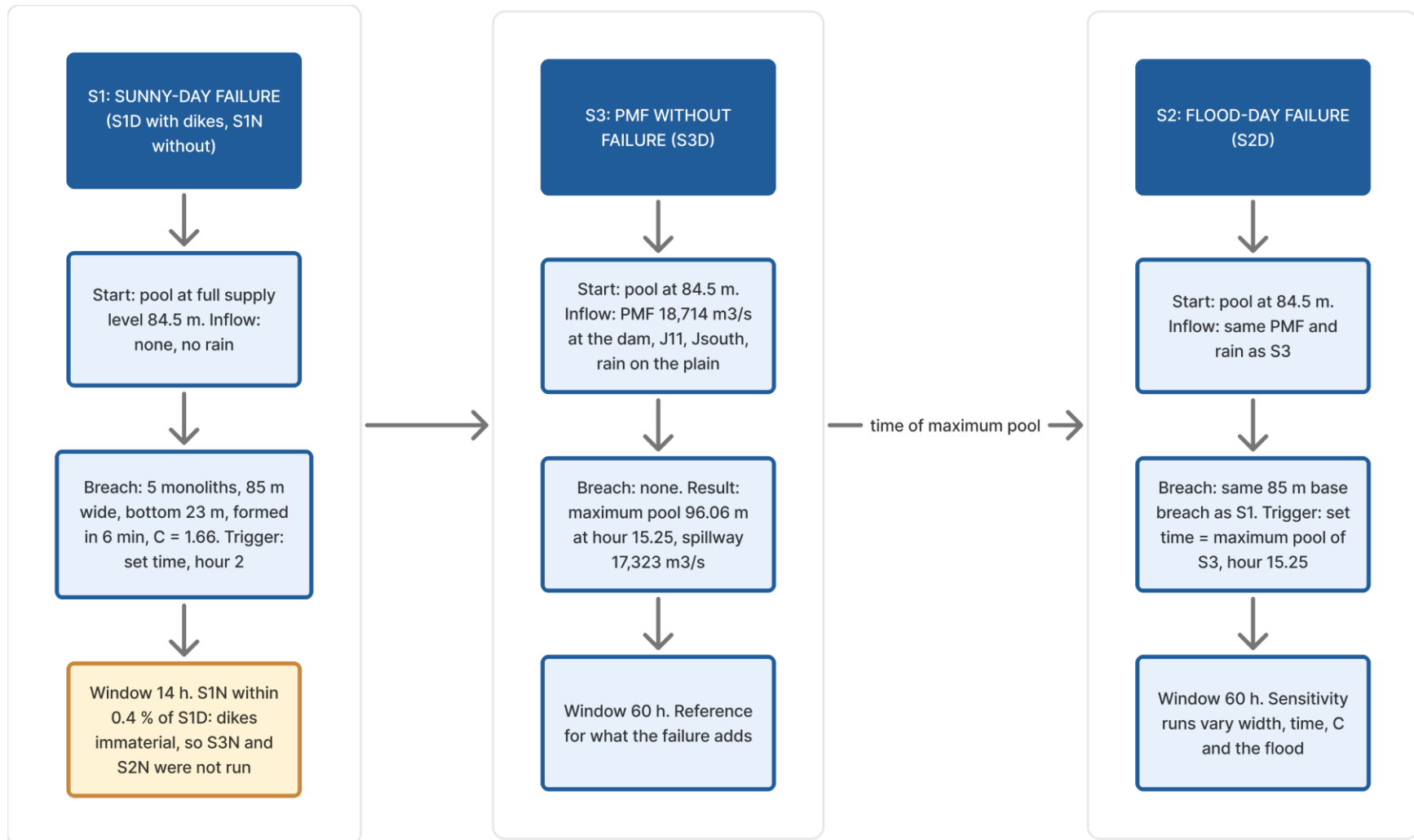


Figure 4 - The three base scenarios and their default conditions.

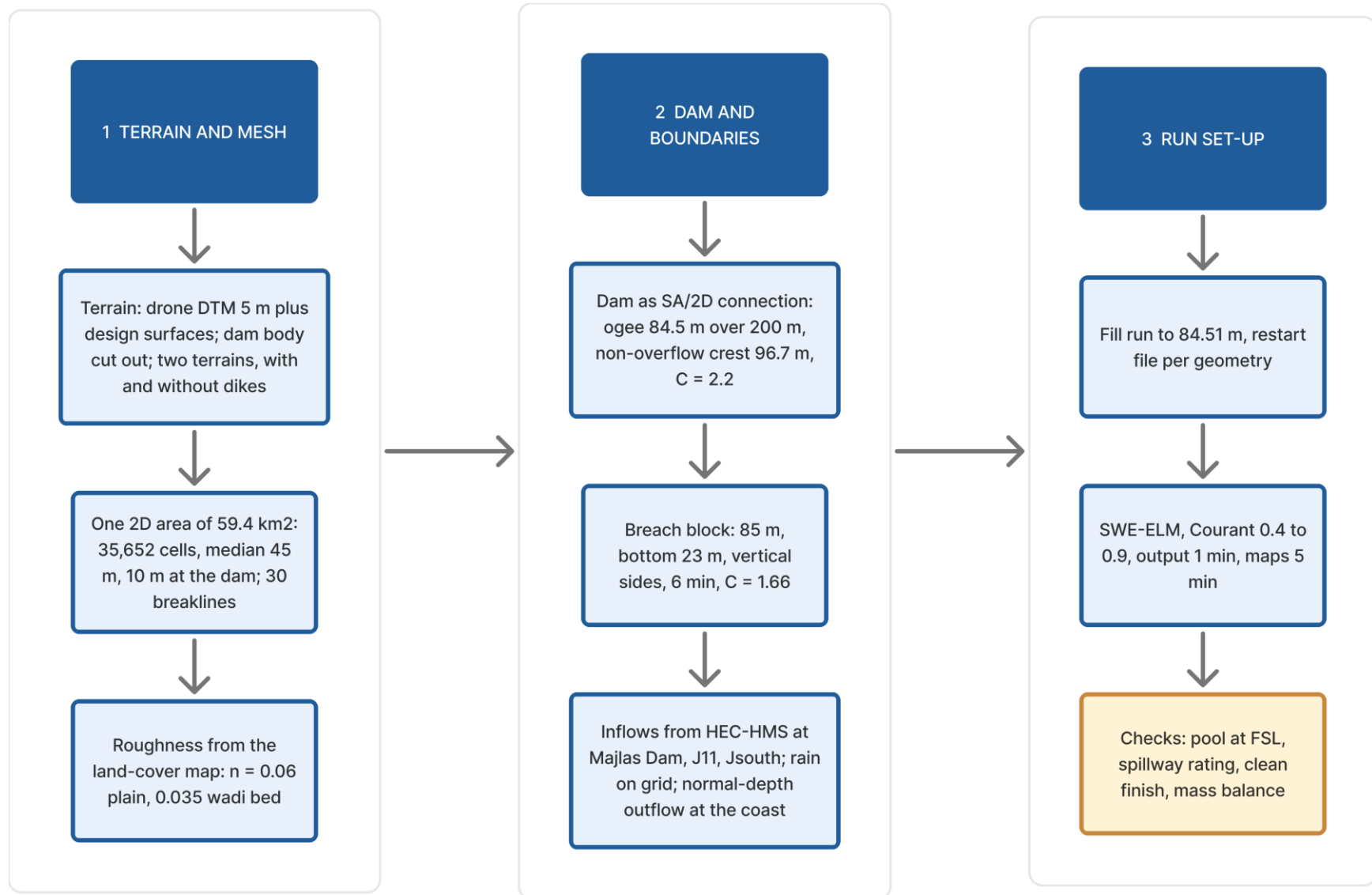


Figure 5 - How the HEC-RAS model was assembled.

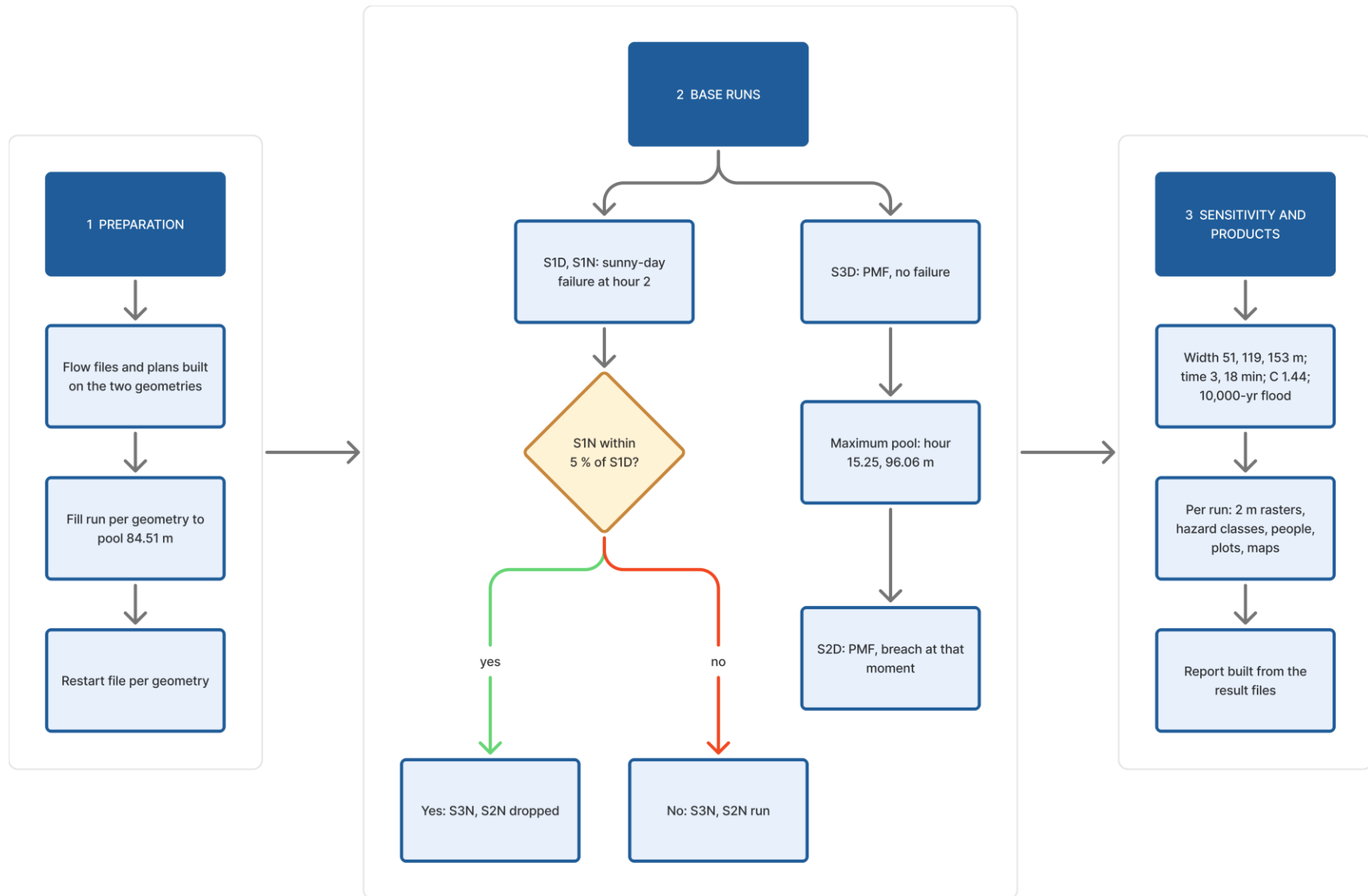


Figure 6 - The run campaign as executed, with the dikes decision gate.



3. SCENARIOS AND RUNS

Three scenarios were run, each with the training dikes in place (D) and without them (N). The names used throughout this report are:

Table 3 - The scenarios.

Name	Scenario	Reservoir at the start	Inflow	Failure
S1D, S1N	Sunny-day failure	Full supply level, 84.5 m	None	Five monoliths at a set time, two hours into the run
S2D, S2N	Flood-day failure	Full supply level, 84.5 m	PMF, with rain on the plain	Five monoliths at the moment the reservoir reaches its maximum level
S3D, S3N	PMF without failure	Full supply level, 84.5 m	PMF, with rain on the plain	None

A sunny-day failure¹ gives the largest breach flow relative to the normal flow and the least warning. The flood-day failure is assumed to happen at the worst moment of the PMF, when the reservoir is at its highest and the load on the dam is greatest; this is the usual convention for a concrete gravity dam, whose critical case is the peak water load rather than overtopping. The PMF without failure is the reference that shows what the failure adds.

3.1. WHY THE NO-DIKES FLOOD-DAY RUNS WERE NOT NEEDED

The training dikes were designed for the 200-year wadi flood. Before running the flood-day pair without them, the two sunny-day runs were compared, because they isolate the dam's contribution with nothing else in the valley. The differences are negligible:

Table 4 - Effect of the training dikes on the sunny-day failure.

Quantity	S1D, with dikes	S1N, without dikes	Difference
Peak flow at the dam (m ³ /s)	45,312	45,346	0.1%
Flooded area > 0.3 m (km ²)	17.0	17.1	0.4%
People in the flooded area	23,212	23,299	0.4%
Median arrival time on the plain	15 min	15 min	0.0%

¹ Sunny-day failure: a failure with no flood in the river, for example from a foundation problem or an earthquake. It gives the least warning, because nobody expects a flood.



A wave of this size overtops the dikes within minutes and spreads over the whole plain; the dikes neither contain it nor steer it to a measurable degree. The flood-day pair without dikes (S3N, S2N) was therefore not run, and the with-dikes runs stand for both conditions. This is a reasoned departure from the methodology report, which had planned both.

3.2. SENSITIVITY RUNS

The sensitivity runs change one breach parameter at a time from the base case: the breach width (three, five, seven and nine monoliths), the formation time (3, 6 and 18 minutes), the weir coefficient of the breach opening, and the flood (the 10,000-year flood instead of the PMF).

Table 5 - The sensitivity runs.

Name	What is changed from the base case	Purpose
S1D-W51, S1D-W153	Breach width 51 and 153 m (three and nine monoliths) instead of 85 m, sunny day	How much the peak and the flooded area depend on how many monoliths go
S2D-W119	Breach width 119 m (seven monoliths), flood day	The same question on the flood day, one check
S1D-T3m, S1D-T18m	Formation time 3 min and 18 min instead of 6 min, sunny day	Whether the speed of the failure matters
S1D-C1.44	Breach weir coefficient 1.44 instead of 1.66, sunny day	Lower bound of the flow through the opening
S3D-F10000, S2D-F10000	10,000-year flood instead of the PMF, without and with failure	How the flood-day results change with a smaller flood

The breach mechanics are tested on the sunny day, where the dam's contribution stands alone and a run takes an hour. The flood-day runs first planned for width (51 and 153 m) and formation time were stopped on 23 September once the sunny-day family had answered the question, S2D-W119 giving the one flood-day check on width, and the machine time went to the 10,000-year pair.



4. THE MODEL AS BUILT

4.1. TERRAIN, MESH AND ROUGHNESS

The model is one two-dimensional area, 'DS Protection', of 59.4 km² in HEC-RAS 6.6, from the upper reservoir to the shoreline. The terrain is the drone DTM 5 m with the design surfaces; the dam body was cut out of it so that the dam connection alone controls the flow. Two terrains and two geometries were built, with the training dikes (Terrain (2).Protection2) and without (Terrain (2)).

Table 6 - The 2D model as built.

Item	As built
Cells	35,652 with dikes, 35,878 without
Cell size	median 45 m, 10 % of cells below 21 m, largest 79 m; 10 m along the dam connection, 25 m beyond
Breaklines	30 lines at 20 to 25 m: dike crests, wadi banks, road embankments, dam axis
Roughness	n = 0.06 (plain, land cover default): 95.8 % of the area; n = 0.035 (wadi bed polygon): 4.2 % of the area
Solver	SWE-ELM, theta 1.0, eddy viscosity 0.3, base step 10 s adapted on Courant 0.4 to 0.9 (floor 0.335 s), output every 1 min, maps every 5 min

4.2. THE DAM AS DESIGNED

The model represents the dam as finalised in the design review of 21 September 2026. It is a roller-compacted concrete (RCC) gravity dam with a grout-enriched RCC facing on a straight axis, 312 m long at the crest and 72.5 m high above the wadi bed (110.5 m above the foundation at -15.0 m a.s.l.). The crest is 6.0 m wide at 95.5 m a.s.l. with a parapet to 96.7 m; the upstream face is 0.05H:1V and the downstream face 0.75H:1V. Figure 7 shows the upstream elevation: 17 monoliths separated by 16 contraction joints, B1 of 31.21 m against the left bank, B2 to B16 of 17.00 m each and B17 of 25.79 m against the right bank. The central blocks under the spillway stand on the foundation at -15.0 m; the abutment blocks step up the banks.

The spillway sits on the central blocks: a free ogee with its crest at 84.5 m a.s.l., the full supply level, in 5 bays of 40 m between 4 piers of 3 m (200 m net, 212 m gross, 218 m between the training walls), a stepped chute at 0.75H:1V and a USBR Type II stilling basin 55 m long with its floor at 15.5 m a.s.l. Its design capacity is 17,000 m³/s. The bottom outlet, one 2.8 m pipe in the left diversion culvert rated 120 m³/s, is closed in every run. Figure 8 and Figure 9 show the non-overflow and spillway sections with their levels; Table 7 lists the values and where each one enters the model.



Table 7 - The dam as designed and where each value enters the model (design review, 21 September 2026).

Item	As designed	In the model
Dam type	Roller-compacted concrete (RCC) gravity dam with grout-enriched RCC facing	Failure by sliding of whole monoliths, vertical breach sides (chapter 2)
Crest / parapet top	95.5 / 96.7 m a.s.l.	Non-overflow crest of the connection at 96.7 m: the parapet is credited, so the PMF pool of 96.06 m does not overtop the dam
Crest width / length	6.0 m / 312 m	Weir width 6 m; connection line along the axis
Height above wadi bed / foundation	72.5 m / 110.5 m	Wadi bed at the axis 23.0 m a.s.l. = breach bottom
Foundation level	-15.0 m a.s.l.	Below the breach bottom; not modelled
Faces, upstream / downstream	0.05H:1V / 0.75H:1V	Vertical breach sides; the chute slope is in the terrain downstream
Monoliths	17 blocks: 31.21 m, 15 × 17.00 m, 25.79 m; 16 joints	Breach width in whole 17 m blocks: 51, 85 (base), 119 and 153 m = 3, 5, 7 and 9 blocks
Spillway	Free ogee at 84.5 m, 5 × 40 m bays, 4 × 3 m piers, net 200 m, 17,000 m ³ /s	Weir profile at 84.5 m over 200 m of the connection; ogee coefficient 2.2
Stilling basin	USBR Type II, 55 m, floor 15.5 m a.s.l.	In the terrain
Bottom outlet	One 2.8 m pipe, intakes 36 and 44 m, 120 m ³ /s	Closed
Reservoir	102.2 Mm ³ at 84.5 m; 150.7 Mm ³ at 96.0 m	Storage from the terrain: 101.7 Mm ³ at 84.51 m after the fill



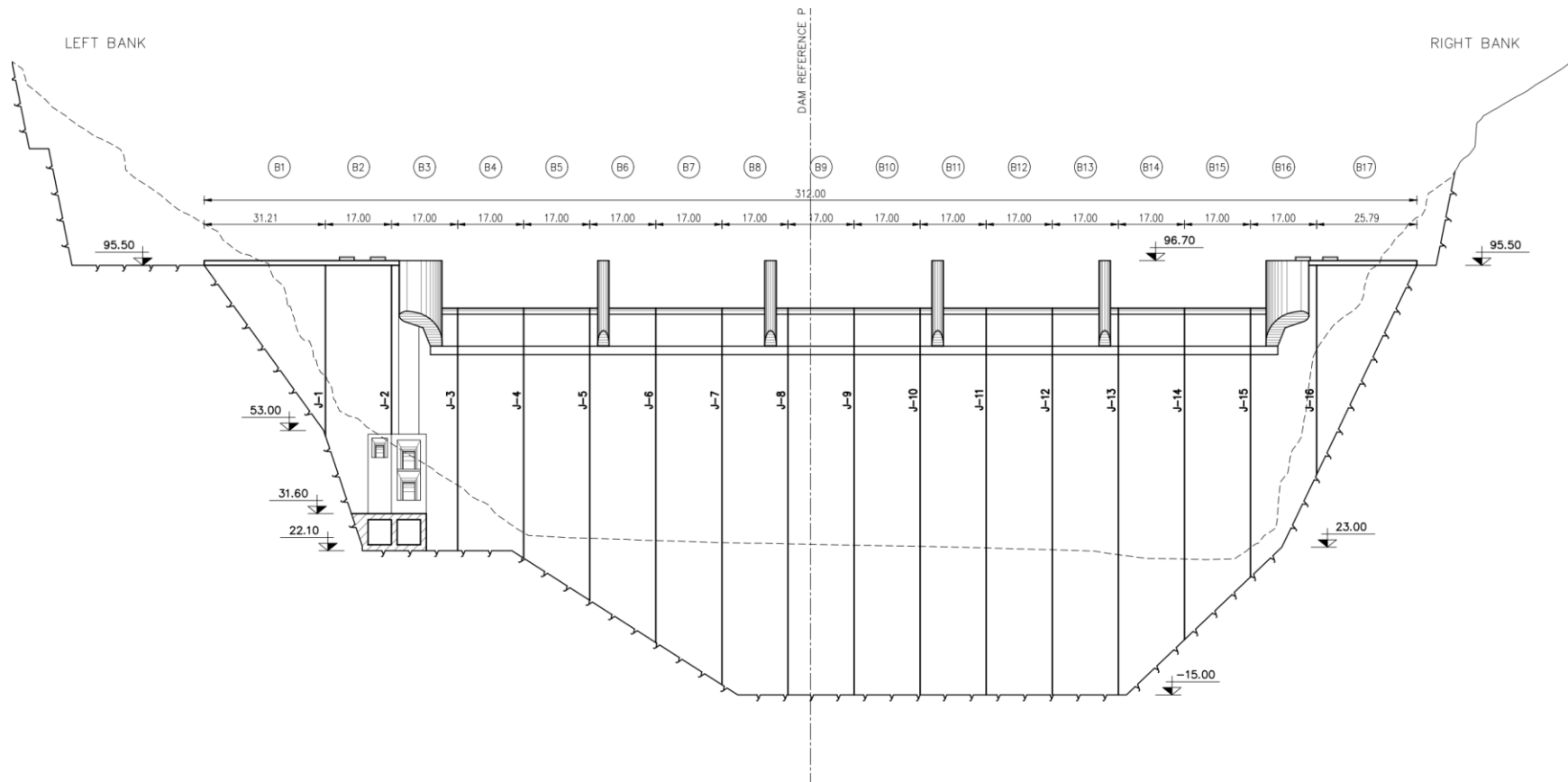


Figure 7 - Upstream elevation of the dam: the 17 monoliths B1 to B17 with their widths, crest 95.5 m, parapet 96.7 m and the foundation at -15.0 m (design review, September 2026).

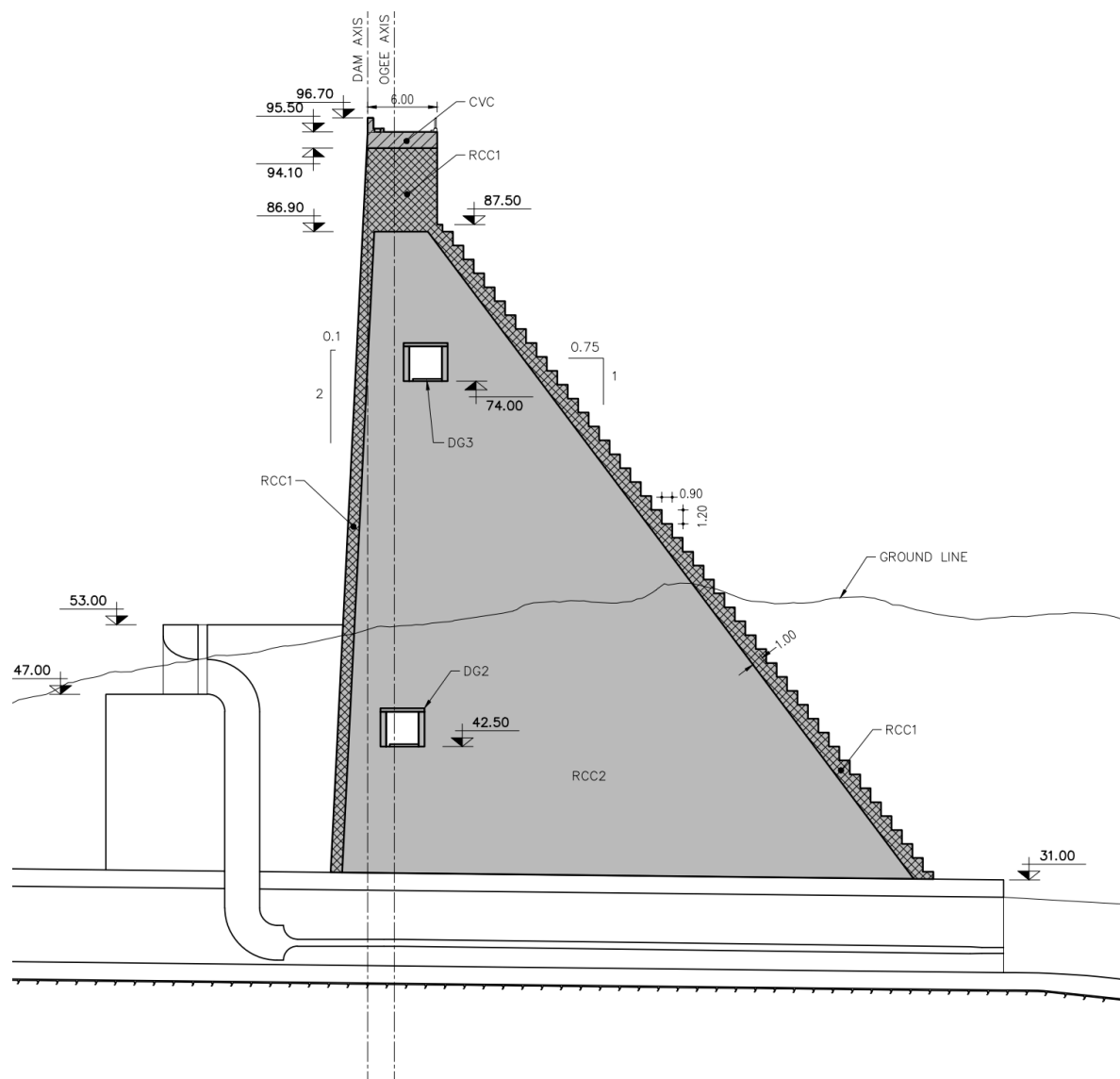


Figure 8 - Section through a non-overflow block: crest 95.5 m, parapet 96.7 m, galleries at 74.0 and 42.5 m, faces 0.05H:1V and 0.75H:1V (design review, September 2026).

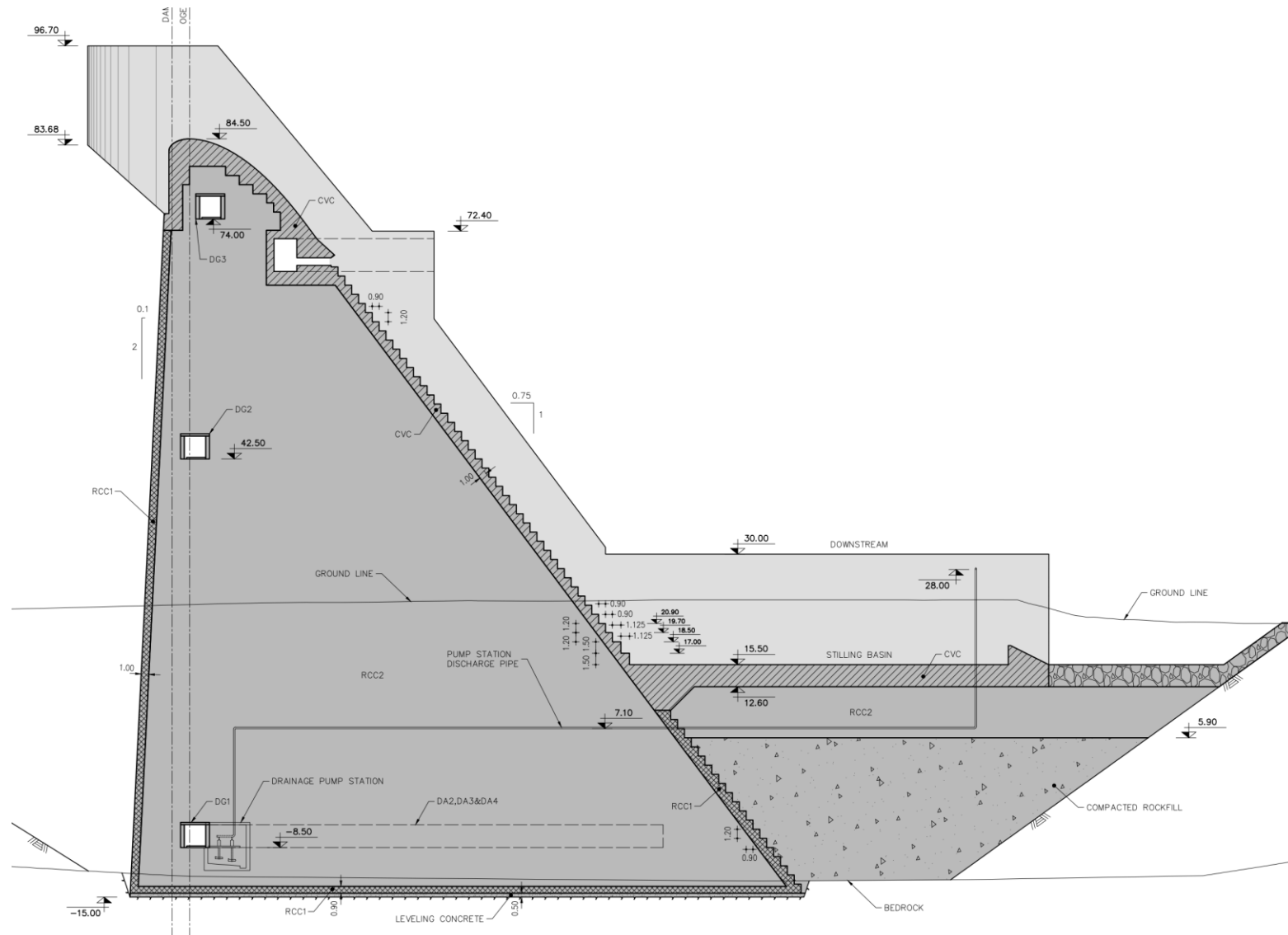


Figure 9 - Section through a spillway block: ogee crest 84.5 m, stepped chute, stilling basin floor 15.5 m, foundation -15.0 m (design review, September 2026).

4.3. THE DAM AND THE BREACH

The dam is a weir connection¹ whose profile carries the non-overflow crest at 96.7 m and the ogee at 84.5 m from station 60 to 260 (200 m); weir coefficient 2.2 (broad crested option), width 6 m, submergence limit 0.98. The breach parameters follow the methodology report:

Table 8 - Breach parameters of the base case.

Parameter	Base value	Basis
Failure mode	Sliding of whole monoliths, vertical sides	Concrete gravity dam; regression equations for earth dams do not apply
Breach width	85 m (five monoliths of 17 m)	Two spillway bays; FERC/USACE guidance: up to half the crest length
Breach bottom	23.0 m a.s.l.	Wadi bed at the axis
Formation time	0.1 h (6 min)	Instantaneous for a concrete dam; the shortest HEC-RAS accepts without instability
Weir coefficient	1.66 (SI)	HEC-RAS range for a broad crest, 1.44 to 1.66
Progression	Sine wave	HEC-RAS option for a fast, smooth opening
Trigger, sunny day	Set time, 2 h into the run	Gives the model two hours to settle on the restart
Trigger, flood day	Set time = moment of maximum pool in the no-failure run	Failure at the highest load; read from S3D or S3N

4.4. BOUNDARY CONDITIONS

Inflow to the reservoir and the flows of the two side wadis joining below the dam (J11 and the southern catchment of 159 km²) come from the HEC-HMS model of the hydrology study, basin model 'Basin 1 Res2', re-run with the southern catchment added. The three elements whose hydrographs are taken are the junction just upstream of the reservoir, J11 and Jsouth. Rain falls directly on the modelled area with the same storm, reduced by the areal factor of 0.76. The sunny-day runs have no inflow and no rain.

¹ SA/2D connection: the HEC-RAS element that links two parts of a 2D area through a structure. Its profile carries the spillway crest and the non-overflow crest; the breach is cut into it.



HEC-HMS basin model 'Basin 1 Res2'

Red elements: hydrographs fed into HEC-RAS | Wadi Majlas Dam Break Analysis | Renardet S.A. & Partners, 2026

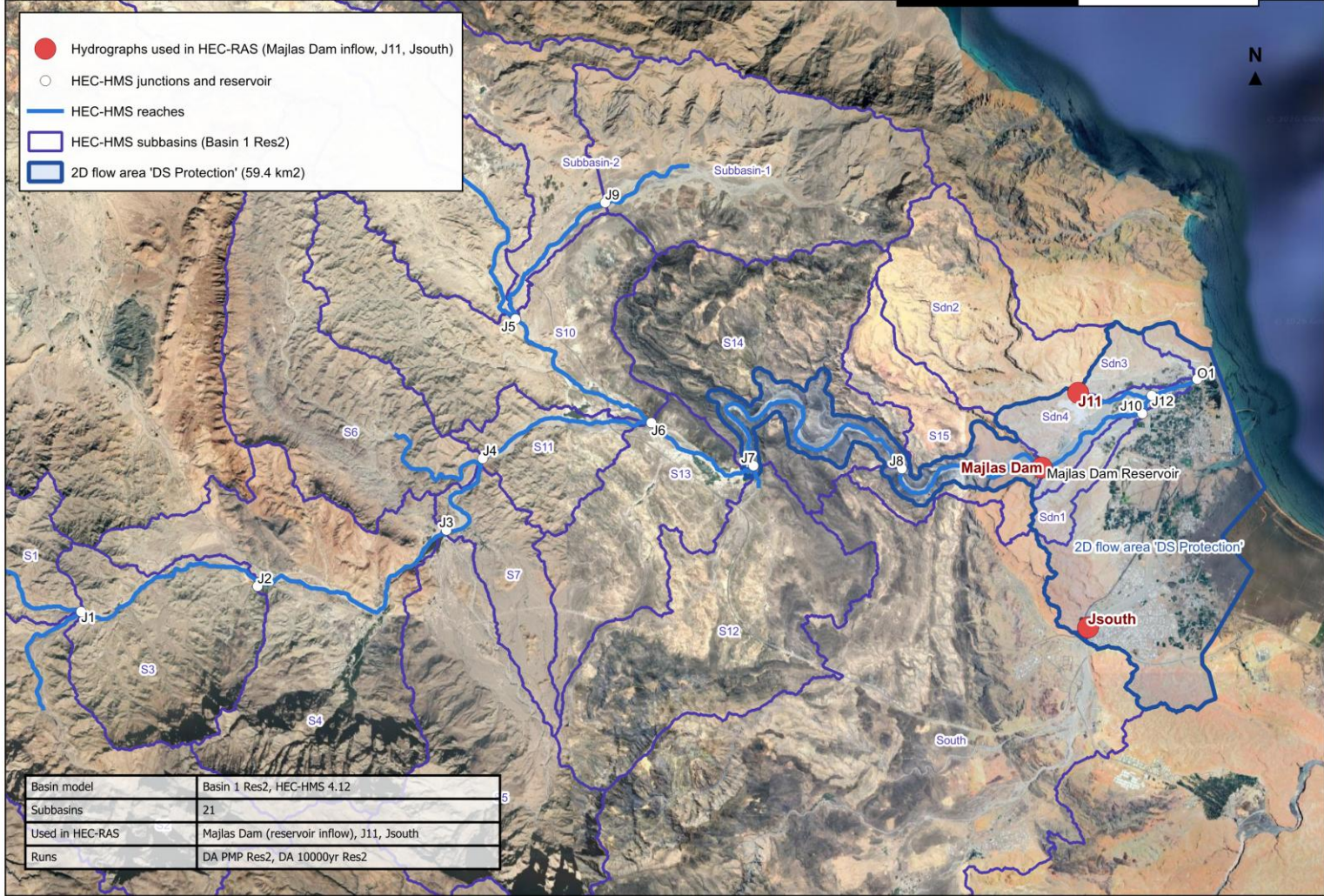


Figure 10 - The HEC-HMS basin model and the three elements whose hydrographs feed the 2D model.



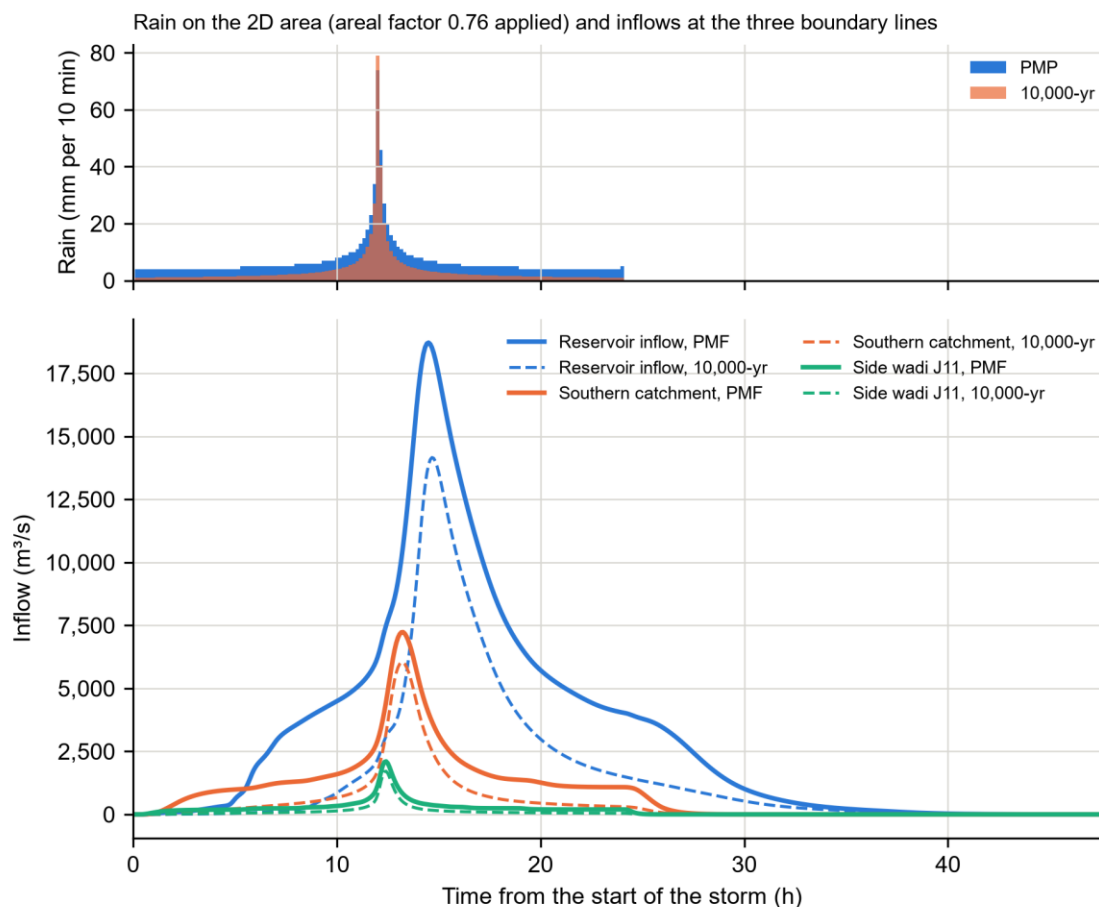


Figure 11 - The inputs to the flood-day runs: rain on the 2D area and the inflow hydrographs at the three boundary lines, PMP and 10,000-year flood.

The numbers were taken from two workbooks kept with the model inputs: 'Hydrology Data.xlsx' holds the three hydrographs and the rain series exactly as entered in HEC-RAS for the PMP and the 10,000-year flood; 'FM rainfall 2-PMP.xlsm' holds the 10-minute rainfall patterns per return period and for the PMP from which those series, with the areal reduction factor, were built.



Table 9 - Boundary conditions of the flood-day runs (times from the start of the storm).

Boundary	PMF	10,000-year flood
Reservoir inflow, 'Majlas Dam' line	18,714 m ³ /s at 14.5 h, 549.4 Mm ³	14,157 m ³ /s at 14.7 h, 278.0 Mm ³
Side wadi J11	2,101 m ³ /s at 12.4 h, 28.4 Mm ³	1,711 m ³ /s at 11.6 h, 12.8 Mm ³
Southern catchment, 'Jsouth' line	7,231 m ³ /s at 13.2 h, 156.9 Mm ³	6,034 m ³ /s at 12.4 h, 81.4 Mm ³
Rain on the 2D area	998 mm, peak 73.9 mm in 10 min	524 mm
Coast, 'Outflow Sea' and 'Outflow 2' lines	normal depth, friction slope 0.003, lines 'Outflow Sea' and 'Outflow 2'	same
Sea level	Mean higher high water at Qurayyat, 2.6 m above chart datum = 0.71 m a.s.l. (chart datum 1.89 m below mean sea level; Oman National Hydrographic Office, Maritime Book 2024); the shoreline terrain is higher, so the sea does not back water into the model	same

4.5. STARTING THE RUNS: THE FILL AND THE RESTART FILE

A run needs the reservoir full at the start. Instead of an initial-condition point, each geometry was filled once by a 48-hour run: 2,500 m³/s for 10.3 hours with one-hour ramps, 101.7 Mm³, which brings the pool to 84.51 m without spilling. The final state was saved as a restart file, and every scenario starts from it. The fills carry the same rain block as the flood runs, with zero values, because HEC-RAS cannot start a run with rain from a restart written by a run without it.



4.6. SOLVER AND DEVIATIONS FROM THE METHODOLOGY

All runs use the shallow-water equations with the Eulerian-Lagrangian scheme (SWE-ELM)¹, an adaptive time step on a Courant number² of 0.4 to 0.9, output every minute and maps every five minutes.

The application deviates from Rev 00 in five points, each for a reason found during the runs:

- Solver: SWE-ELM instead of the SWE-EM named in the methodology, for the stability reason given in the footnote.
- Non-overflow crest: the parapet (96.7 m) is credited as the overtopping level, as on the design drawings.
- Flood-day trigger: set at the time of maximum pool of the no-failure twin rather than at a fixed level, so that a PMF peaking below the parapet still triggers the failure.
- Initial state: fill run and restart file instead of an initial-condition point; the result is the same start.
- Flood-day sensitivity runs use a 36-hour window (the breach falls between hour 15 and 20) instead of 60 hours, the sunny-day runs 14 hours (breach at hour 2); the base flood-day runs keep 60 hours.

¹ HEC-RAS offers two full momentum solvers. SWE-ELM is the faster and more robust one; SWE-EM keeps more of the momentum terms but proved unstable on this mesh at the first spill over the dry chute.

² Courant number: the distance the water moves in one time step divided by the cell size. Keeping it below one keeps the solution stable.



5. RESULTS: THE PMF WITHOUT FAILURE

The no-failure runs show what the valley faces from the PMF alone, and give the maximum reservoir level and the moment of the failure for the flood-day runs.

Table 10 - PMF without failure: key results.

Scenario	Description	Peak flow at dam (m ³ /s)	Time of peak	Max pool (m a.s.l.)	Area > 0.3 m (km ²)	Max depth gorge / plain (m)	People in flooded area
S3D	PMF without failure, dikes in place	17,323	15 h 15 min	96.06	25.9	47 / 12.2	44,088
S3N	PMF without failure, no dikes	not run, see chapter 3	-	-	-	-	-
S3D-F10000	10,000-year flood without failure, dikes in place	12,386	15 h 30 min	93.75	24.8	46 / 12.0	39,083

With the dikes in place the reservoir peaks at 96.06 m a.s.l., 0.56 m above the crest and 0.64 m below the parapet, and the spillway passes 17,323 m³/s against a design discharge of 17,000 m³/s.



S3D: maximum depth

S3D PMF no failure, dikes | Wadi Majlas Flood Protection Dam, Dam Break Analysis | Renardet S.A. & Partners, 2026

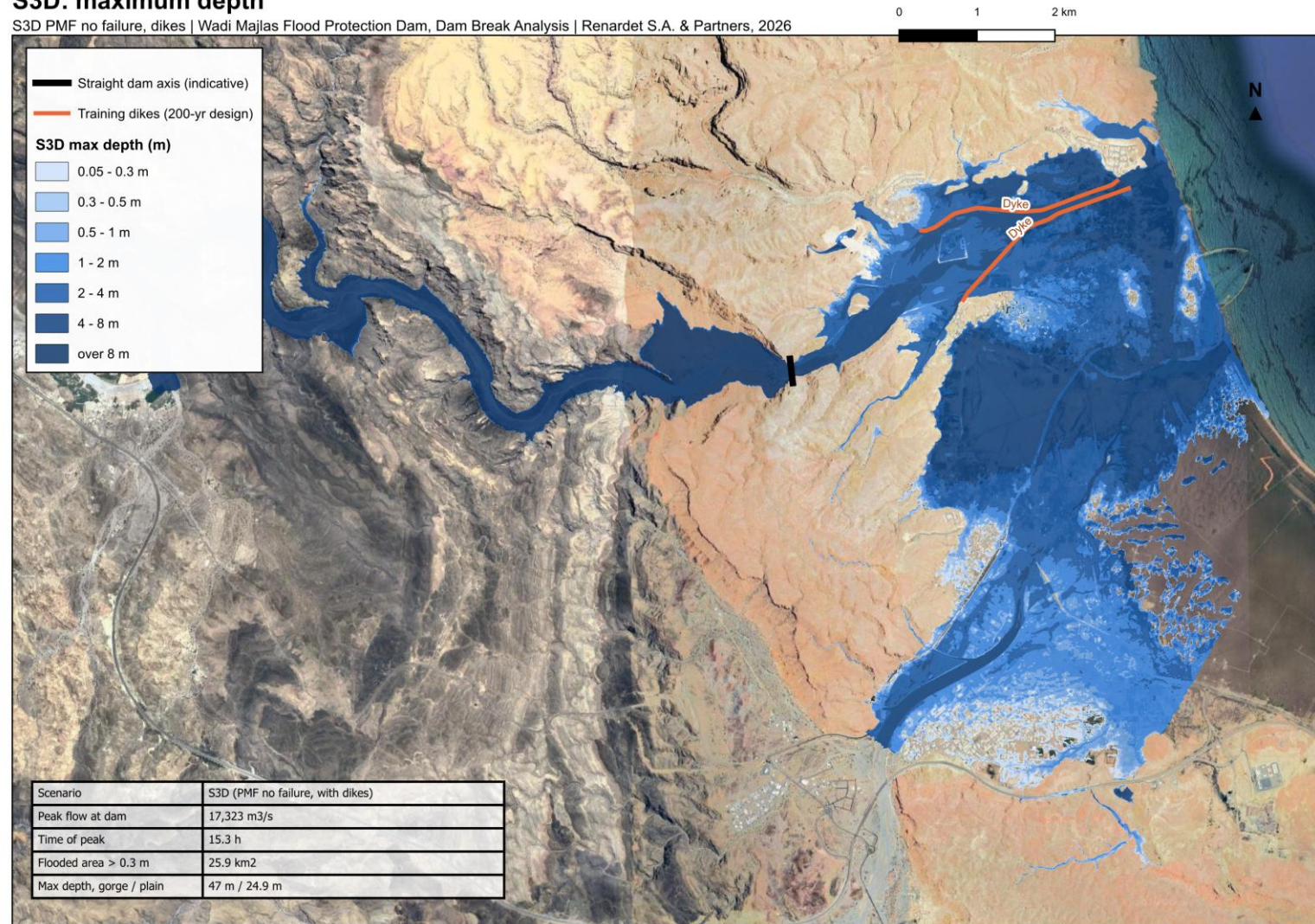


Figure 12 - S3D, PMF without failure, dikes in place: maximum depth.

6. RESULTS: SUNNY-DAY FAILURE

The sunny-day runs answer the question the emergency plan asks first: how much water, how fast, and where, when the dam fails with no flood in the wadi.

Table 11 - Sunny-day failure: key results.

Scenario	Description	Peak flow at dam (m ³ /s)	Time of peak	Max pool (m a.s.l.)	Area > 0.3 m (km ²)	Max depth gorge / plain (m)	People in flooded area
S1D	sunny-day failure, dikes in place	45,312	2 h 10 min	84.51	17.0	54 / 11.5	23,212
S1N	sunny-day failure, no dikes	45,346	2 h 10 min	84.51	17.1	54 / 11.3	23,299

The breach opens two hours into the run. Within six minutes the opening is 85 m wide down to the wadi bed, and the outflow peaks at 45,312 m³/s. The reservoir level falls from 84.51 m to below 40 m a.s.l. within one hour and 103 Mm³ leave the reservoir in about two hours. The wave fills the gorge to more than 40 m depth at the dam, reaches the coastal plain within 15 min of the breach, the shoreline within 22 min and the last corner of the flooded area after 1 h 25 min. In the gorge below the dam the water stands 54 m deep; on the plain depths reach 11.5 m and velocities 12.7 m/s.



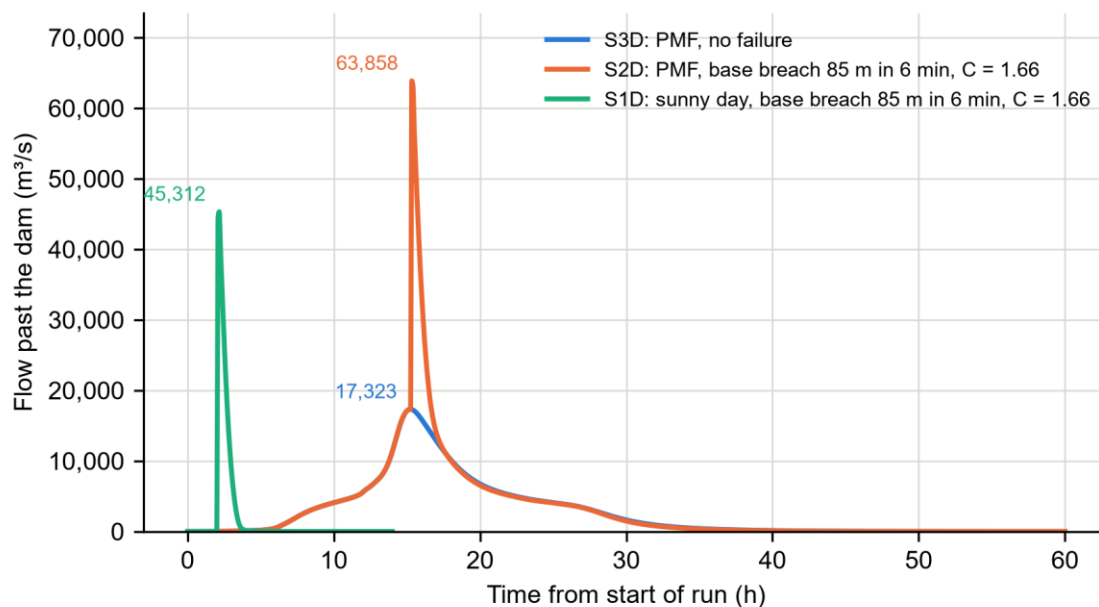


Figure 13 - Flow past the dam, base runs with dikes.

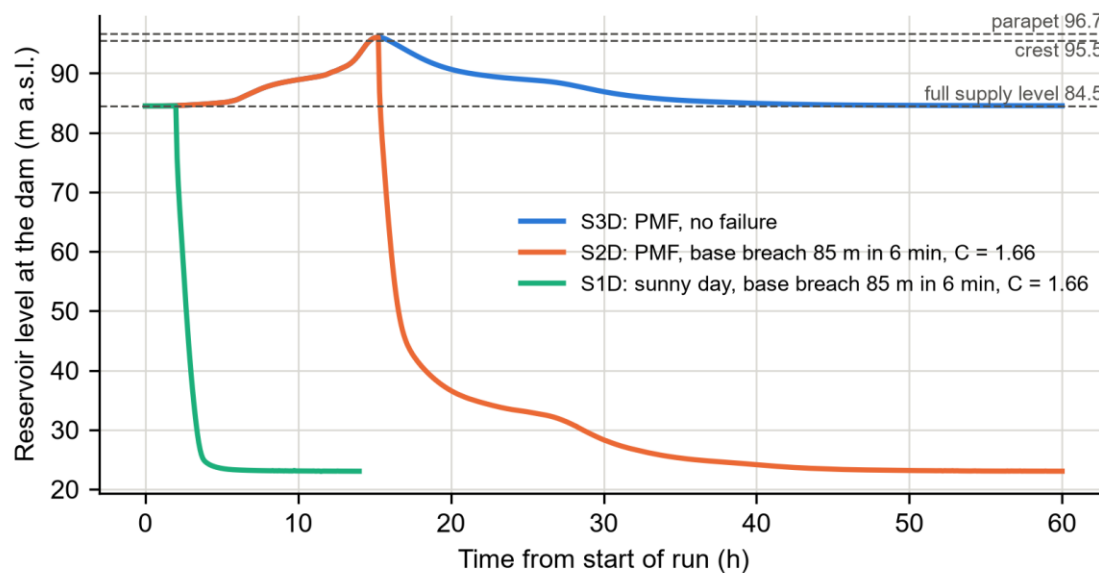


Figure 14 - Reservoir level at the dam, base runs.

S1D: maximum depth

S1D sunny day failure, dikes | Wadi Majlas Flood Protection Dam, Dam Break Analysis | Renardet S.A. & Partners, 2026

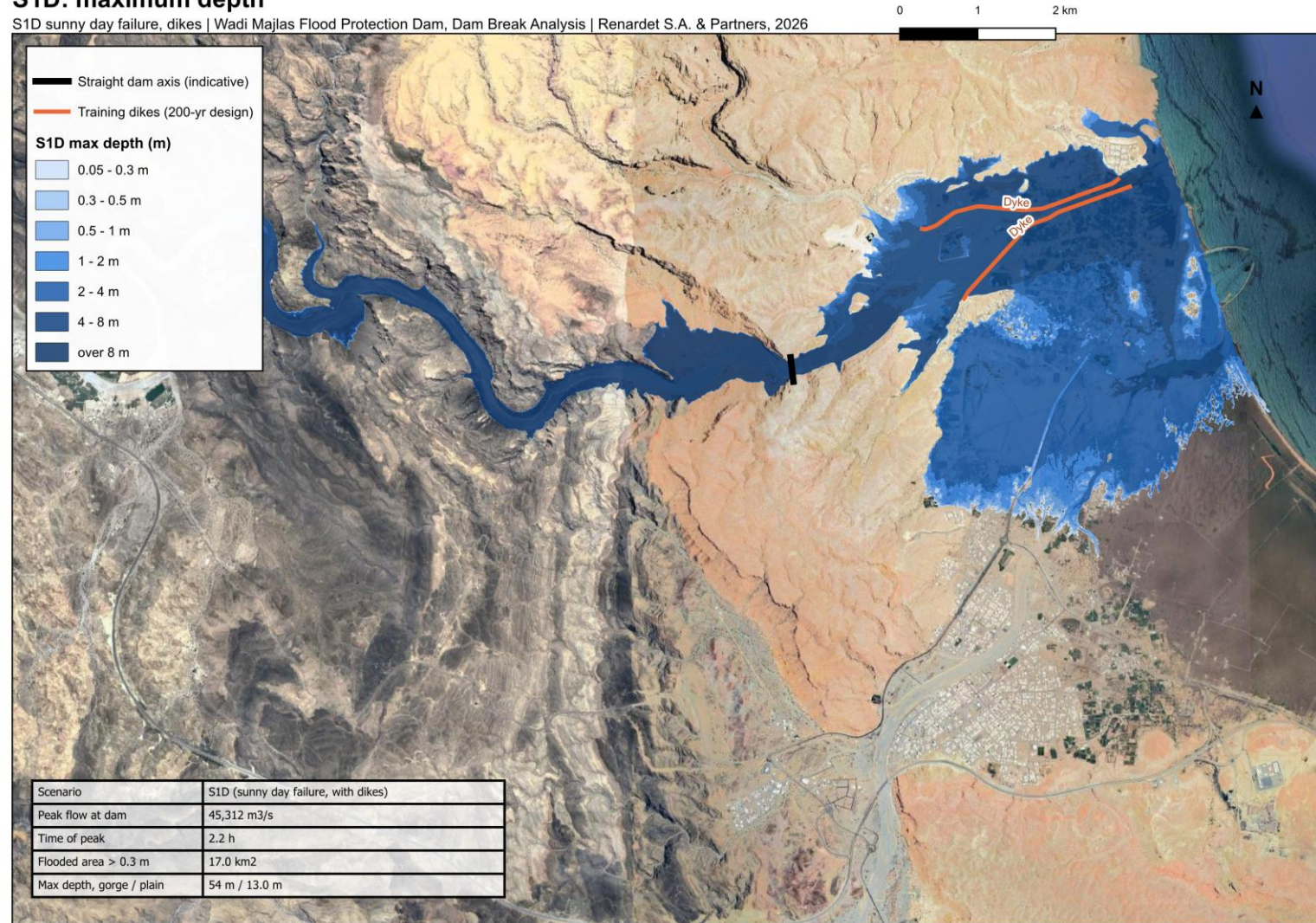


Figure 15 - S1D, sunny-day failure, dikes in place: maximum depth.

S1D: arrival time of a 0.3 m depth after the breach

S1D sunny day failure, dikes | Wadi Majlas Flood Protection Dam, Dam Break Analysis | Renardet S.A. & Partners, 2026

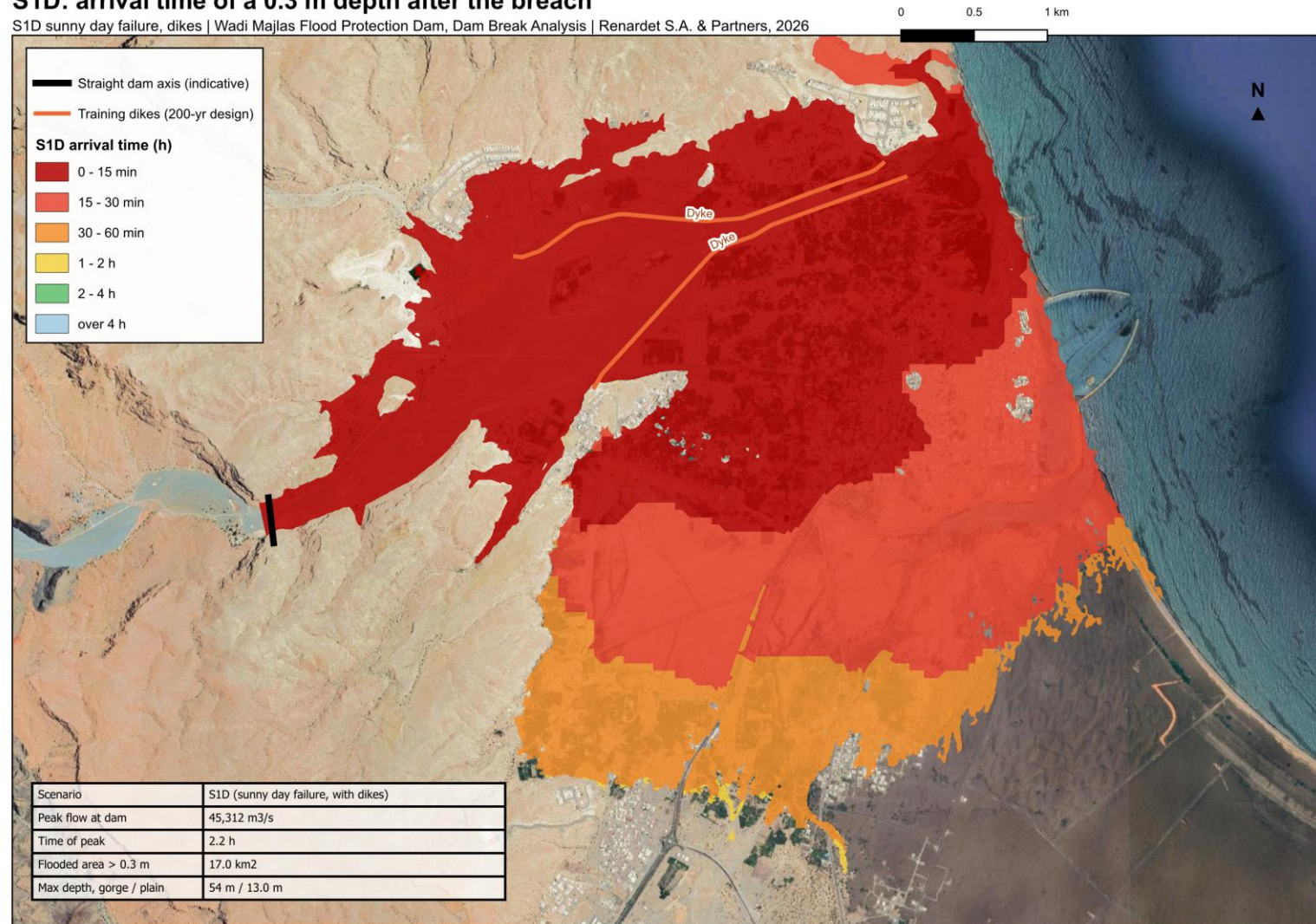


Figure 16 - S1D: arrival time of a 0.3 m depth after the breach, Qurayat.

S1D: flood hazard class (AIDR H1 to H6)

S1D sunny day failure, dikes | Wadi Majlas Flood Protection Dam, Dam Break Analysis | Renardet S.A. & Partners, 2026

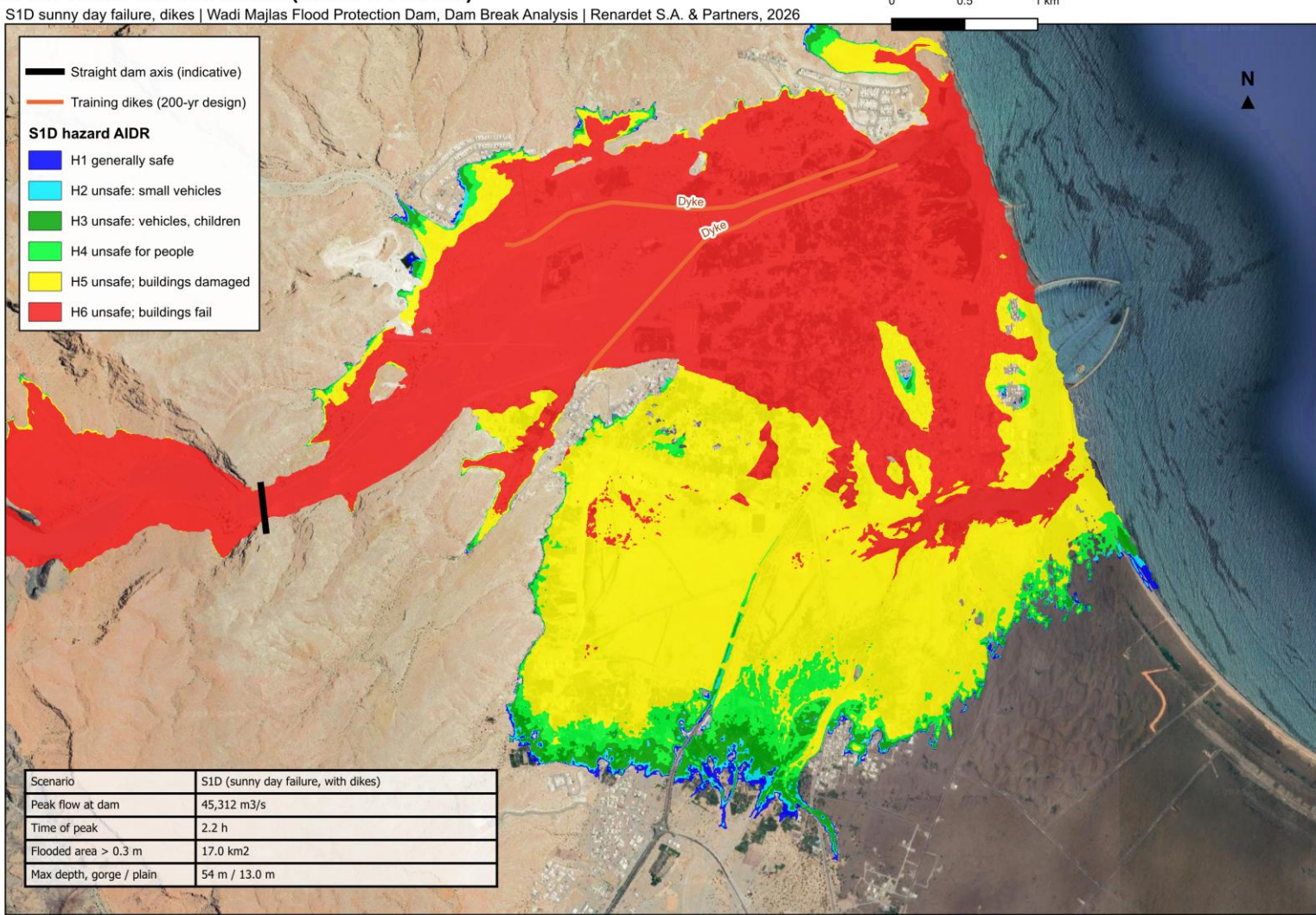


Figure 17 - S1D: flood hazard class (AIDR), Qurayat.

S1N: maximum depth

S1N sunny day failure, no dikes | Wadi Majlas Flood Protection Dam, Dam Break Analysis | Renardet S.A. & Partners, 2026

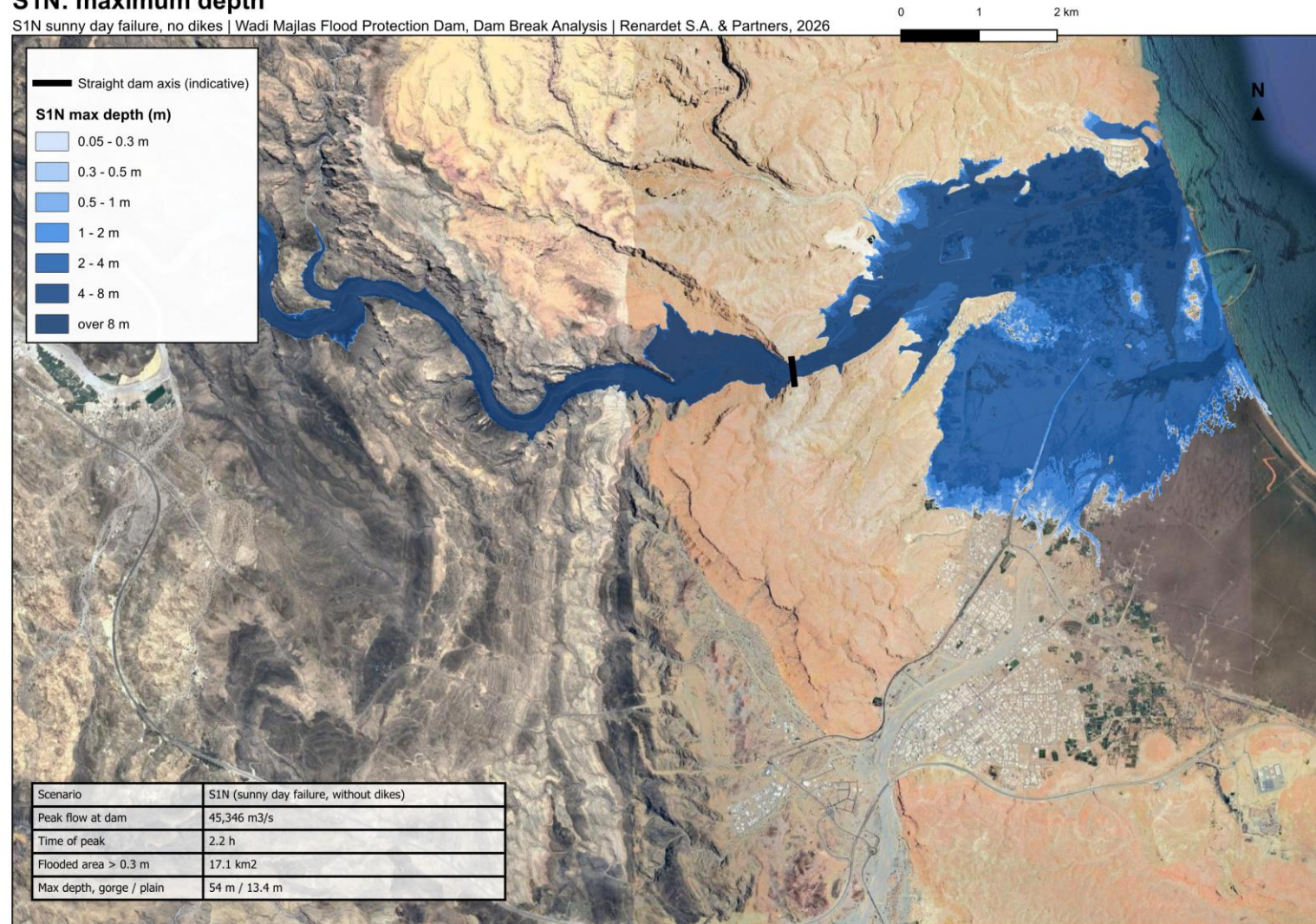


Figure 18 - S1N, sunny-day failure, no dikes: maximum depth.

S1N: arrival time of a 0.3 m depth after the breach

S1N sunny day failure, no dikes | Wadi Majlas Flood Protection Dam, Dam Break Analysis | Renardet S.A. & Partners, 2026

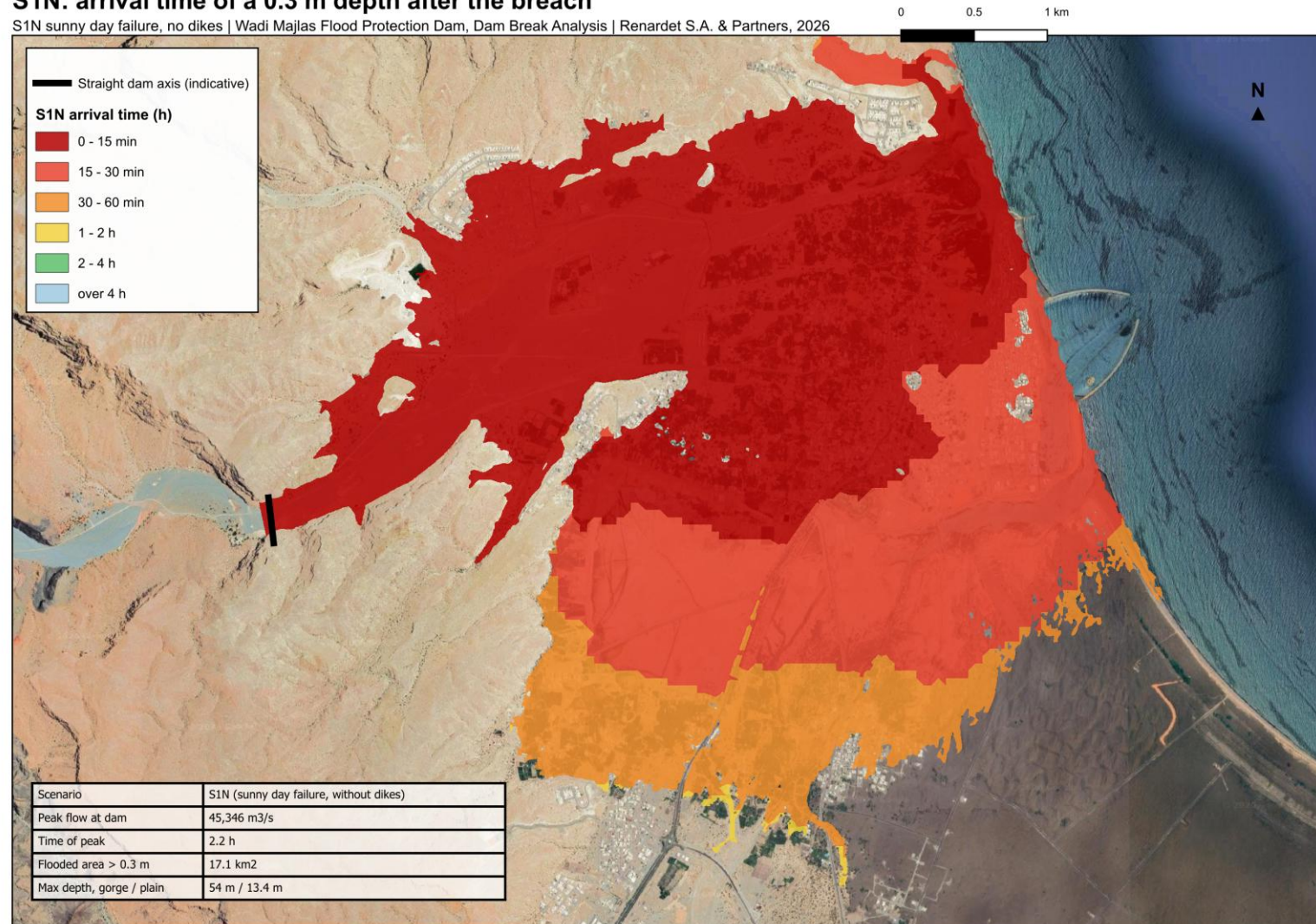


Figure 19 - S1N: arrival time of a 0.3 m depth after the breach, Qurayat.

S1N: flood hazard class (AIDR H1 to H6)

S1N sunny day failure, no dikes | Wadi Majlas Flood Protection Dam, Dam Break Analysis | Renardet S.A. & Partners, 2026

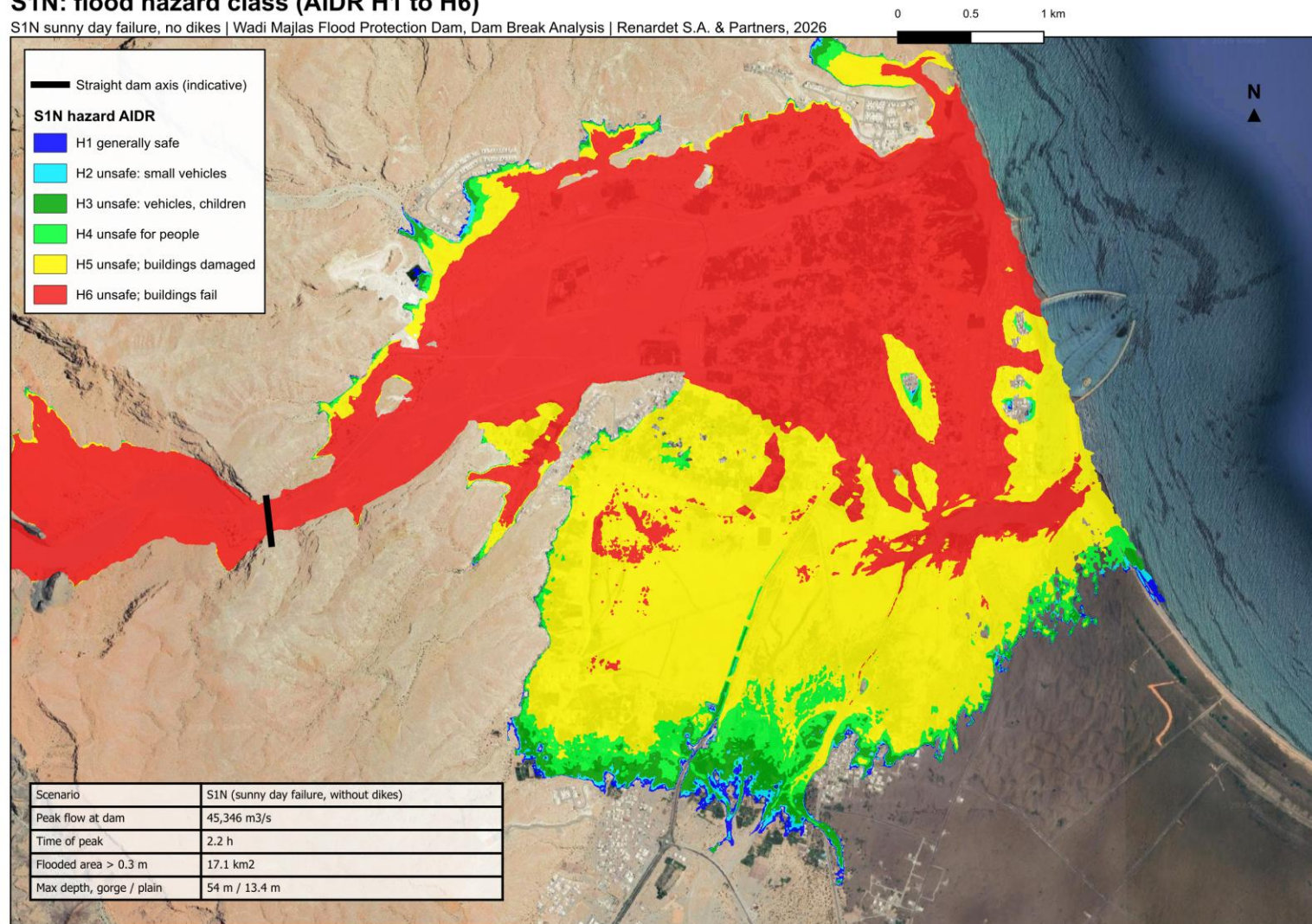


Figure 20 - S1N: flood hazard class (AIDR), Qurayat.

7. RESULTS: FLOOD-DAY FAILURE

The flood-day runs add the failure to a valley already under the PMF; the difference to the no-failure run of chapter 5 is the dam's contribution.

Table 12 - Flood-day failure: key results.

Scenario	Description	Peak flow at dam (m ³ /s)	Time of peak	Max pool (m a.s.l.)	Area > 0.3 m (km ²)	Max depth gorge / plain (m)	People in flooded area
S2D	flood-day (PMF) failure, dikes in place	63,858	15 h 20 min	96.06	27.1	65 / 13.5	46,299
S2N	flood-day (PMF) failure, no dikes	not run, see chapter 3	-	-	-	-	-
S2D-F10000	flood-day failure during the 10,000-year flood	pending	-	-	-	-	-

The failure is triggered 15 h 15 min into the storm, when the reservoir stands at 96.06 m a.s.l. The outflow peaks at 63,858 m³/s, against 17,323 m³/s over the spillway without failure. The flooded area grows only from 25.9 to 27.1 km², because the PMF has already covered the plain; what the failure adds is depth and force. The area in the worst hazard class, H6, grows from 13.0 to 18.4 km², and the people in the flooded area from 44,088 to 46,299. The plain is under water before the dam fails, so the arrival that matters is that of the failure wave on top of the flood: chapter 9 times the moment the failure adds 0.3 m to the depth of the no-failure run and counts the people it reaches.



S2D: maximum depth

S2D flood day PMF failure, dikes | Wadi Majlas Flood Protection Dam, Dam Break Analysis | Renardet S.A. & Partners, 2026

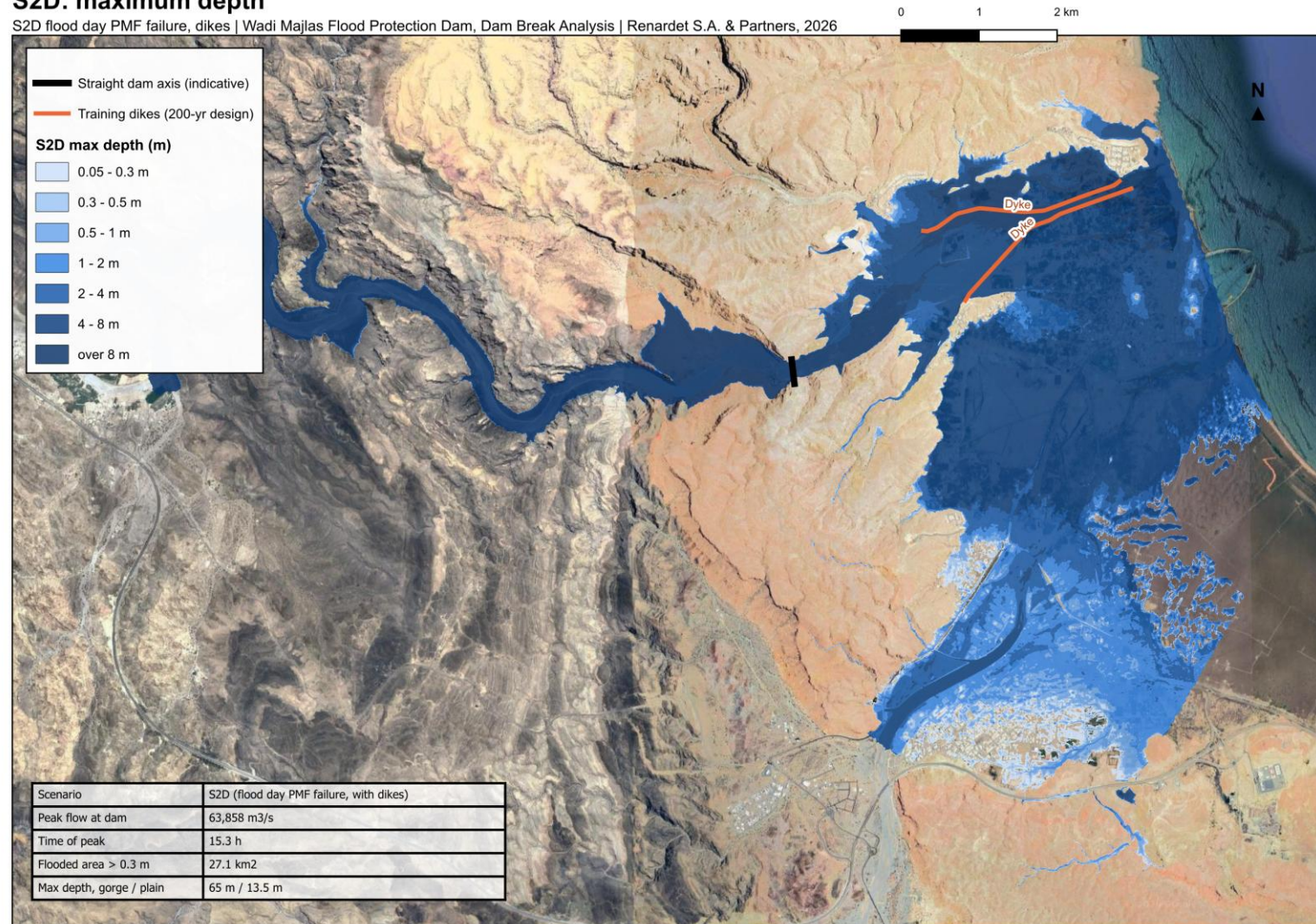


Figure 21 - S2D, flood-day (PMF) failure, dikes in place: maximum depth.

S2D: flood hazard class (AIDR H1 to H6)

S2D flood day PMF failure, dikes | Wadi Majlas Flood Protection Dam, Dam Break Analysis | Renardet S.A. & Partners, 2026

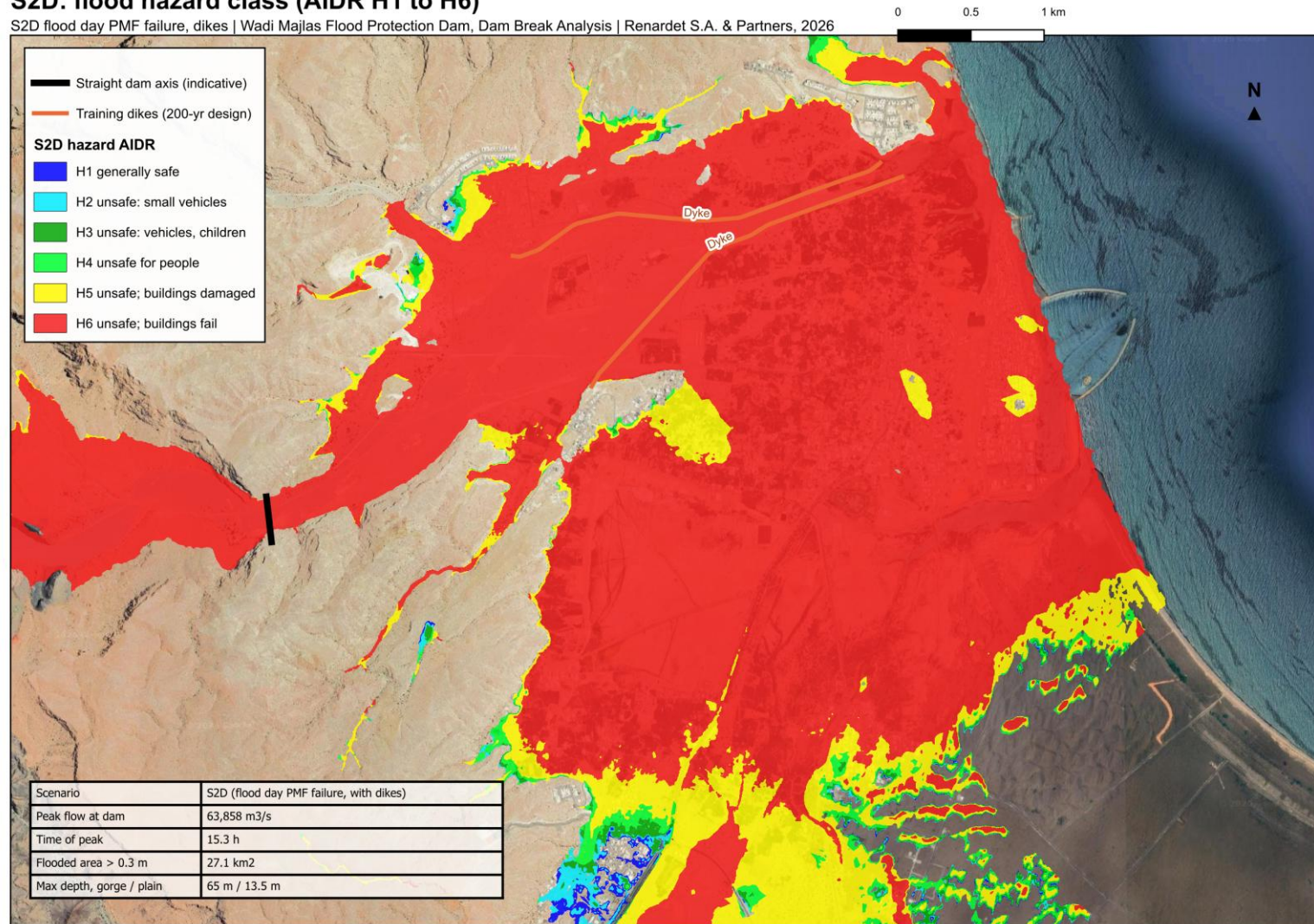


Figure 22 - S2D: flood hazard class (AIDR), Qurayat.

8. SENSITIVITY OF THE RESULTS TO THE BREACH PARAMETERS

The breach is prescribed, not computed, so its width, speed and discharge coefficient were varied one at a time to see how far the results move.

Table 13 - Sensitivity runs: key results.

Scenario	Description	Peak flow at dam (m ³ /s)	Time of peak	Max pool (m a.s.l.)	Area > 0.3 m (km ²)	Max depth gorge / plain (m)	People in flooded area
S1D-W51	sunny-day failure, breach 51 m wide	31,663	2 h 10 min	84.51	16.4	52 / 10.2	21,999
S1D-W153	sunny-day failure, breach 153 m wide	66,955	2 h 05 min	84.51	17.4	56 / 13.0	24,148
S1D-C1.44	sunny-day failure, breach weir coefficient 1.44	42,384	2 h 10 min	84.51	16.9	52 / 11.2	22,876
S1D-T3m	sunny-day failure, breach formed in 3 minutes	49,525	2 h 05 min	84.51	17.0	55 / 11.5	23,212
S1D-T18m	sunny-day failure, breach formed in 18 minutes	42,856	2 h 20 min	84.51	17.0	53 / 11.5	23,211
S2D-W119	flood-day failure, breach 119 m wide (7 monoliths)	79,030	15 h 20 min	96.06	27.5	66 / 21.3	46,960
S3D-F10000	10,000-year flood without failure, dikes in place	12,386	15 h 30 min	93.75	24.8	46 / 12.0	39,083
S2D-F10000	flood-day failure during the 10,000-year flood	pending	-	-	-	-	-
S2D-W51	flood-day failure, breach 51 m wide (3 monoliths)	not run, see chapter 3	-	-	-	-	-
S2D-W153	flood-day failure, breach 153 m wide (9 monoliths)	not run, see chapter 3	-	-	-	-	-
S2D-T18m	flood-day failure, breach formed in 18 minutes	stopped at 36 %, see chapter 3	-	-	-	-	-
S2D-T3m	flood-day failure, breach formed in 3 minutes	stopped at 24 %, see chapter 3	-	-	-	-	-



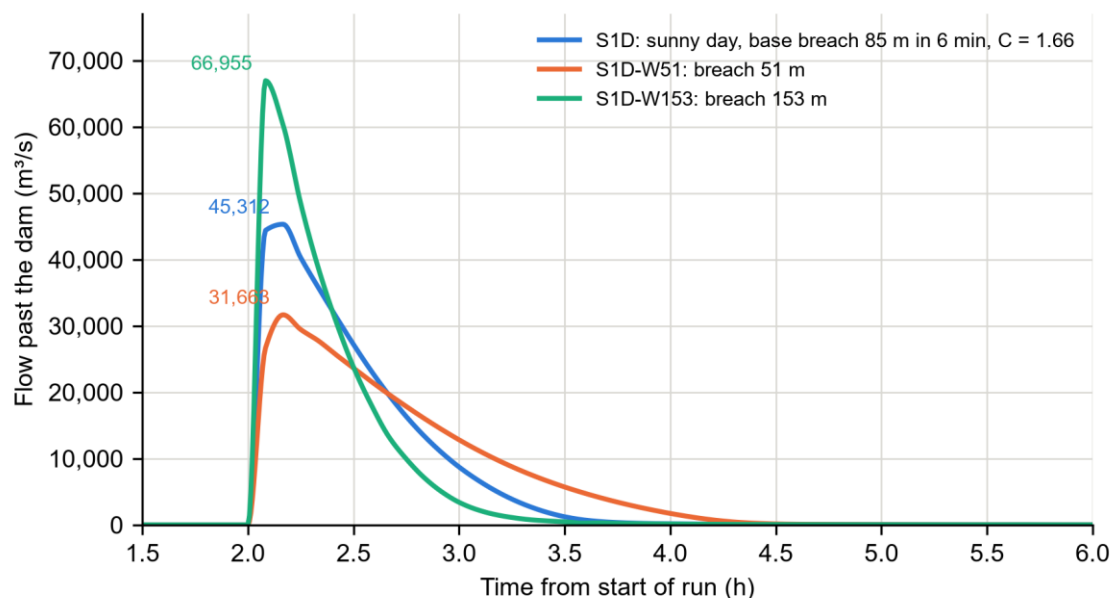


Figure 23 - Flow past the dam for the sunny-day breach-width runs (51, 85 and 153 m).

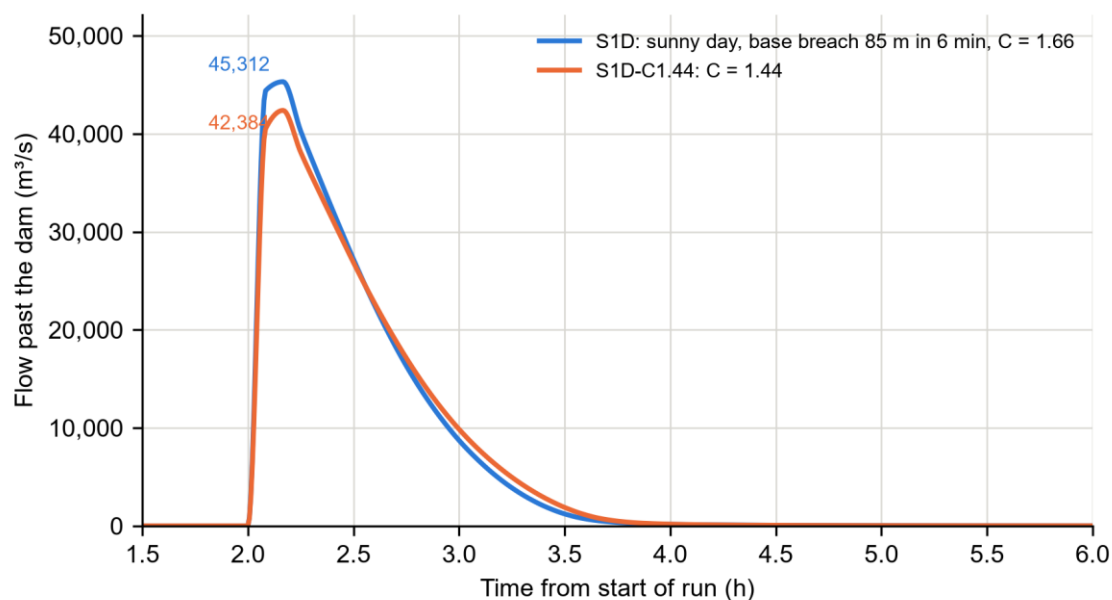


Figure 24 - Flow past the dam for the two breach weir coefficients, sunny day.

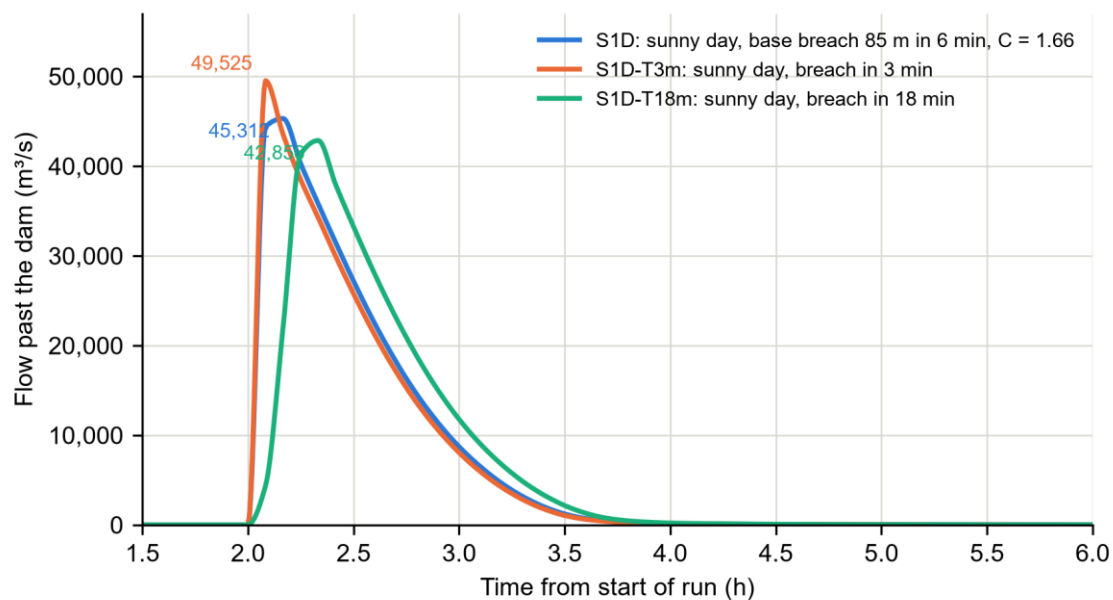


Figure 25 - Flow past the dam for the three formation times, sunny day (3, 6 and 18 min).

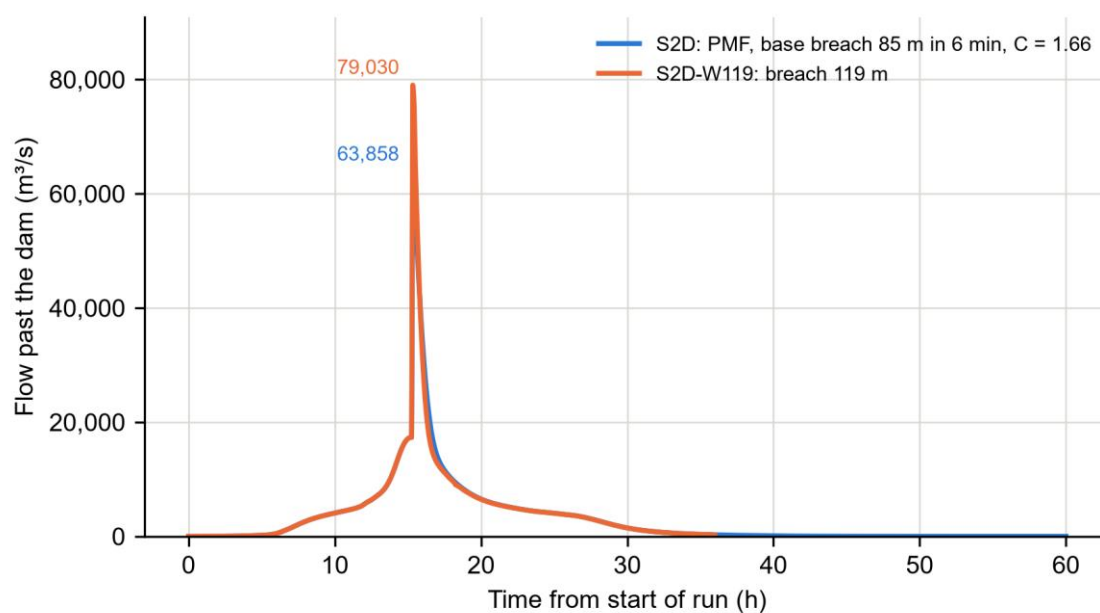


Figure 26 - Flow past the dam for the flood-day breach-width check (85 and 119 m).

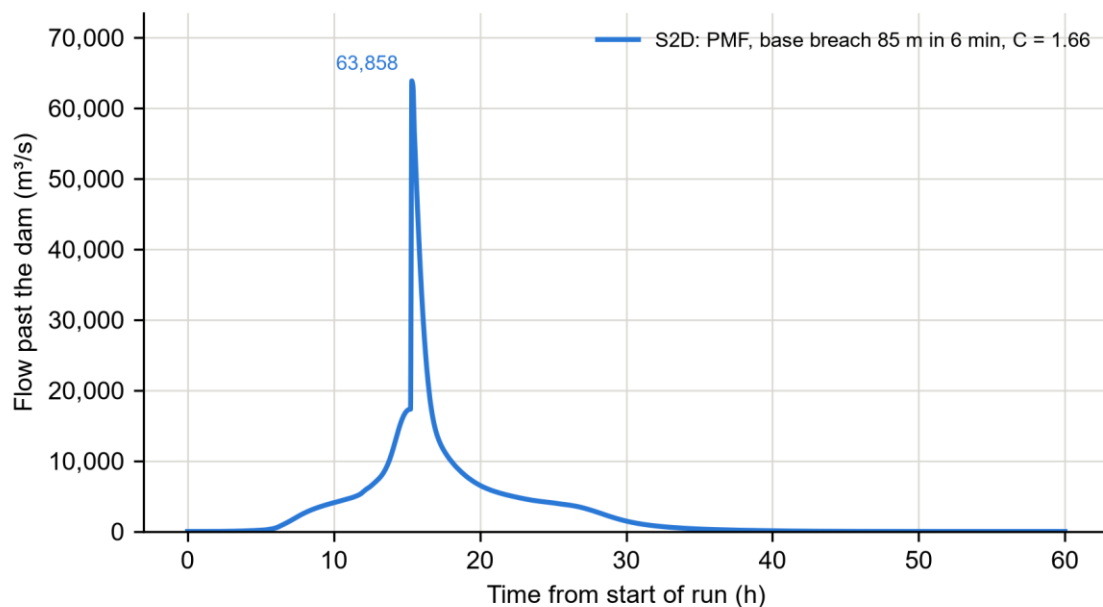


Figure 27 - Flow past the dam for the flood-day failure during the PMF and during the 10,000-year flood.

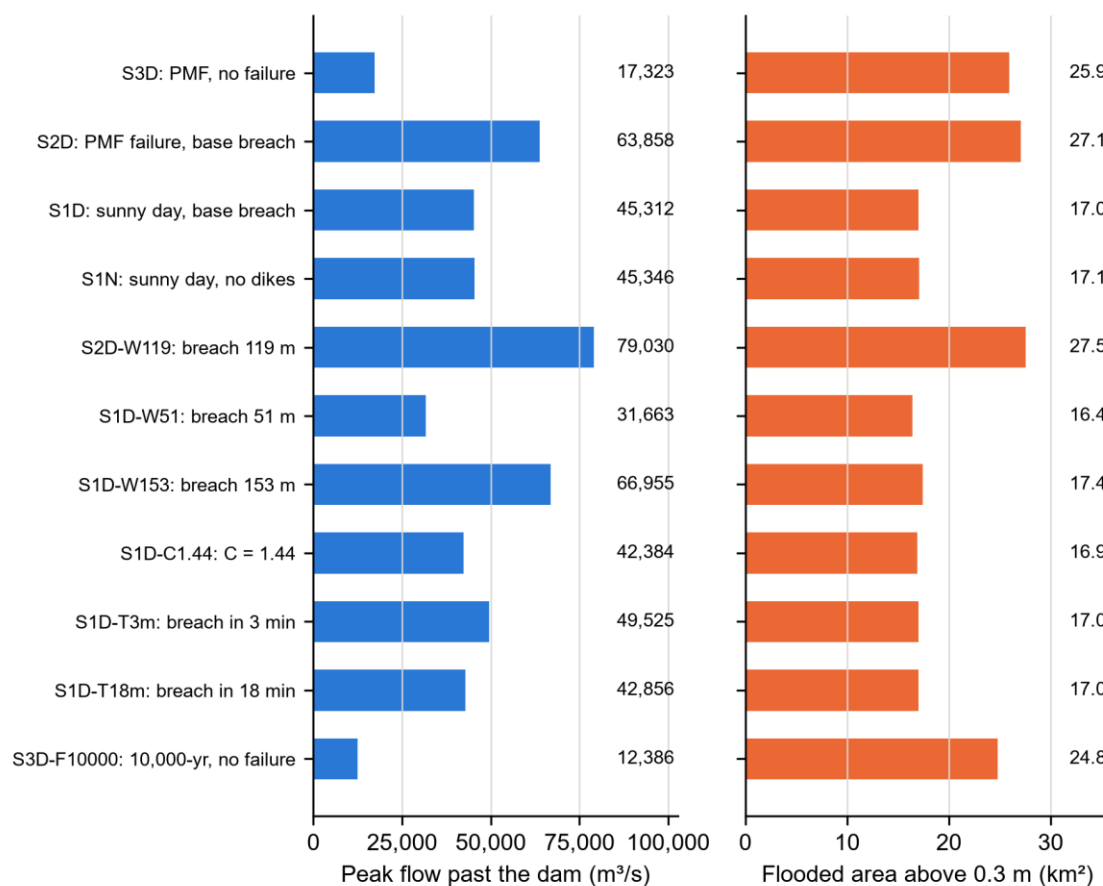


Figure 28 - Peak flow and flooded area of all runs.



The reading of these runs is in chapter 10. In short: the breach width sets the peak almost in proportion (31,663, 45,312 and 66,955 m³/s for 51, 85 and 153 m) while the flooded area hardly moves (16.4 to 17.4 km²), because the plain is wide and flat. The formation time changes the peak by about a tenth either way (49,525 m³/s in 3 minutes, 42,856 in 18) and neither the extent nor the people in it, but it does change the first quarter hour: 18,050 people are reached within 15 minutes of a 3-minute breach, 15,579 with 6 minutes and 4,871 with 18. The lower weir coefficient trims the peak by 6 %.



9. CONSEQUENCES

People and land use in the flooded area were counted on the population grid¹ and the plot layer of the flood-risk study. The hazard classes follow the Australian flood hazard curves (AIDR), which combine depth and velocity into six classes from H1, safe for people and vehicles, to H6, where buildings fail.

Table 14 - People, plots and hazard classes per scenario.

Scenario	People in flooded area	of whom in H4 or worse	Plots touched	of which residential	H1 to H3 (km ²)	H4 (km ²)	H5 (km ²)	H6 (km ²)
S3D	44,088	40,121	3,065	1,777	2.2	1.9	10.1	13.0
S1D	23,212	22,541	2,950	1,677	0.9	0.9	6.8	8.7
S1N	23,299	22,655	2,963	1,687	0.9	0.9	6.7	8.9
S2D	46,299	42,911	3,341	2,038	1.8	1.4	6.6	18.4
S1D-W51	21,999	21,491	2,806	1,557	1.0	1.2	7.1	7.4
S1D-W153	24,148	23,819	3,036	1,756	0.8	0.8	5.6	10.5
S1D-C1.44	22,876	22,394	2,936	1,668	0.9	0.9	6.9	8.5
S1D-T3m	23,212	22,541	2,950	1,677	0.9	0.9	6.8	8.7
S1D-T18m	23,211	22,540	2,950	1,677	0.9	0.9	6.8	8.7
S2D-W119	46,960	43,451	3,404	2,099	1.9	1.3	6.6	19.0
S3D-F10000	39,083	35,537	2,780	1,572	2.8	2.7	12.1	8.7

¹ GHS-POP 2025, a global population grid of 100 m cells from the European Commission; the count is the sum of the cells whose centre lies in water deeper than 0.3 m.



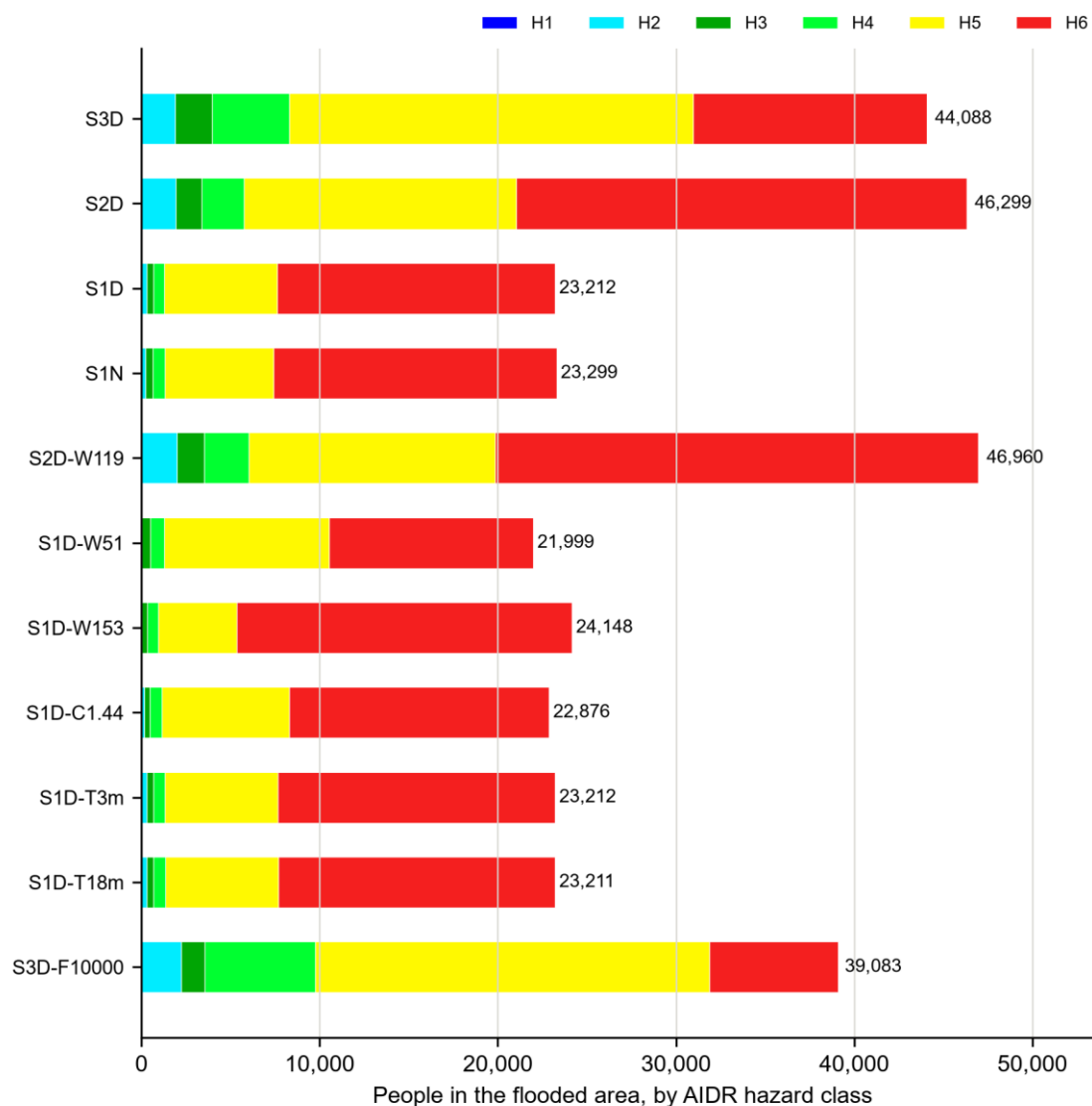


Figure 29 - People in the flooded area by hazard class, per scenario.

9.1. WARNING TIME: WHO IS REACHED, AND WHEN

The hazard maps alone make the PMF without failure look almost as bad as the failure: similar area, similar classes. What separates them is time. The sunny-day wave reaches the town within minutes of the breach; the PMF takes hours to build up, and its rain and side wadis wet the plain long before the reservoir spills. This section puts the scenarios on the clock that matters for the emergency plan.

Each scenario is timed from its own trigger: a failure from the moment the breach opens, the PMF without failure from the start of the storm. A place counts as reached when the water there rises 0.3 m above what was there before the trigger. For the flood-day failure the plain is already under the



PMF when the dam goes, so it is timed against the no-failure run at the same instant: the moment the failure makes the water 0.3 m deeper than the flood alone would. The model gives travel times; the warning time available to people is shorter by the time it takes to detect the failure and to spread the alarm, which depend on the dam's instrumentation and the authorities' procedures and are not set by this study.

Table 15 gives the time at which the rise reaches five places along the wadi, from the gorge exit to the shore; the flood-day columns are the failure wave over the PMF, the S3D column counts from the start of the storm.

Table 15 - Arrival of a 0.3 m rise at five places: failures timed from the breach, S3D from the start of the storm.

Place	S1D	S1N	S1D-W51	S1D-W153	S2D	S2D-W119	S3D
Gorge exit, start of the fan	5 min	5 min	5 min	5 min	< 5 min	< 5 min	40 min
Head of the training dikes	5 min	5 min	5 min	5 min	< 5 min	< 5 min	43 min
Qurayat, residential centre	15 min	10 min	15 min	10 min	10 min	5 min	1 h 06 min
Shoreline, 'Outflow Sea' line	22 min	22 min	30 min	20 min	10 min	10 min	1 h 38 min
Southern outlet, 'Outflow 2' line	not reached	not reached	not reached	not reached	55 min	45 min	2 h 57 min

Table 16 counts, for the failures, the people and the plots reached within each time after the breach; the last column is everyone in the flooded area, whether or not the rise reaches 0.3 m. Figure 30 draws the same counts as curves.



Table 16 - People and plots reached within a given time after the breach (cumulative).

Scenario	within 15 min	within 30 min	within 45 min	within 1 h	within 1 h 30 min	within 2 h	within 3 h	Total
S1D, people	15,579	21,739	22,706	23,179	23,179	23,179	23,179	23,212
S1D, plots	1,944	2,493	2,831	2,949	2,950	2,950	2,950	2,950
S1N, people	15,765	21,822	22,899	23,266	23,267	23,267	23,267	23,299
S1N, plots	1,991	2,494	2,844	2,961	2,963	2,963	2,963	2,963
S1D-W51, people	7,586	20,715	20,945	21,740	21,966	21,966	21,966	21,999
S1D-W51, plots	1,267	2,368	2,480	2,726	2,802	2,802	2,802	2,802
S1D-W153, people	21,948	22,994	24,083	24,116	24,116	24,116	24,116	24,148
S1D-W153, plots	2,459	2,695	3,019	3,035	3,036	3,036	3,036	3,036
S1D-C1.44, people	13,866	21,427	22,251	22,795	22,844	22,844	22,844	22,876
S1D-C1.44, plots	1,842	2,481	2,776	2,930	2,936	2,936	2,936	2,936
S1D-T3m, people	18,050	21,786	22,845	23,179	23,179	23,179	23,179	23,212
S1D-T3m, plots	2,147	2,511	2,847	2,950	2,950	2,950	2,950	2,950
S1D-T18m, people	4,871	21,697	22,176	23,108	23,178	23,178	23,178	23,211
S1D-T18m, plots	693	2,463	2,661	2,913	2,950	2,950	2,950	2,950
S2D, people	25,365	26,129	26,887	31,500	31,631	31,936	31,936	46,299
S2D, plots	3,250	3,331	3,341	3,341	3,341	3,341	3,341	3,341
S2D-W119, people	25,787	26,787	30,089	33,466	33,746	34,175	34,175	46,960
S2D-W119, plots	3,305	3,403	3,404	3,404	3,404	3,404	3,404	3,404

Table 17 does the same for the flood without failure, in hours from the start of the storm.

Table 17 - People and plots reached within a given time after the start of the storm, no failure (cumulative).

Scenario	within 3 h	within 6 h	within 9 h	within 12 h	within 15 h	within 18 h	Total
S3D, people	11,921	27,933	38,020	44,017	44,056	44,056	44,088
S3D, plots	1,258	2,200	2,754	3,057	3,065	3,065	3,065
S3D-F10000, people	359	5,682	18,314	37,473	39,051	39,051	39,083
S3D-F10000, plots	26	704	1,600	2,702	2,780	2,780	2,780



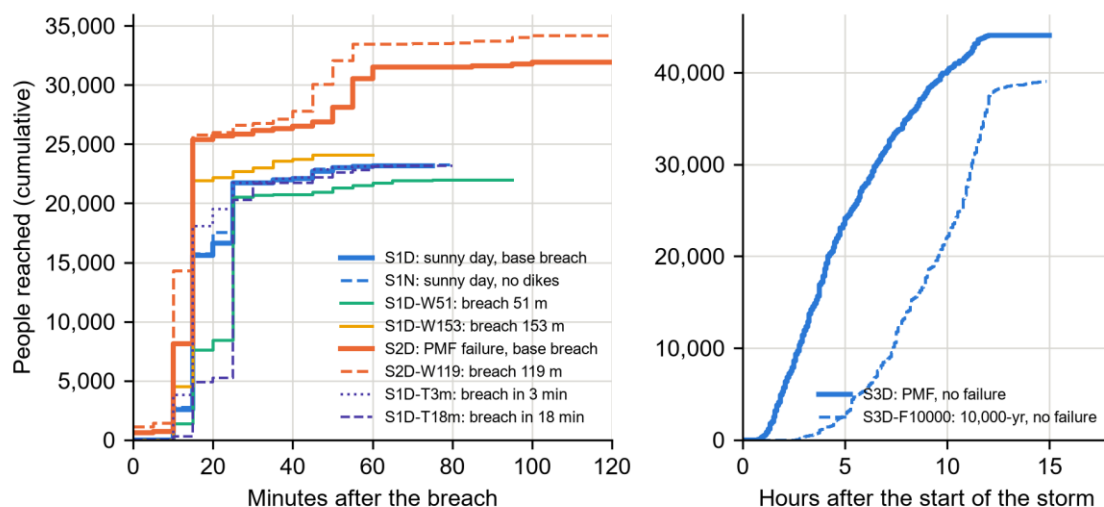


Figure 30 - People reached against time: failures from the breach (left), the PMF without failure from the start of the storm (right).

Table 18 splits the people reached in each interval by the hazard class they end up in, the two inputs a life-safety estimate needs: how little warning, and how severe the flood. H4 and worse is unsafe for people on foot.

Table 18 - People newly reached in each interval after the breach, by final hazard class: sunny-day and flood-day failure.

Reached within	S1D H1-H3	S1D H4	S1D H5	S1D H6	S2D H1-H3	S2D H4	S2D H5	S2D H6
15 min	52	53	3,242	12,233	101	117	3,411	21,736
15 min to 30 min	161	215	2,447	3,337	362	116	276	11
30 min to 45 min	142	186	639	0	191	92	347	128
45 min to 1 h	315	158	0	0	10	31	3,347	1,226
1 h to 1 h 30 min	0	0	0	0	0	0	127	5
1 h 30 min to 2 h	0	0	0	0	0	0	98	206



S1D: people in the flooded area

S1D sunny day failure, dikes | Wadi Majlas Flood Protection Dam, Dam Break Analysis | Renardet S.A. & Partners, 2026

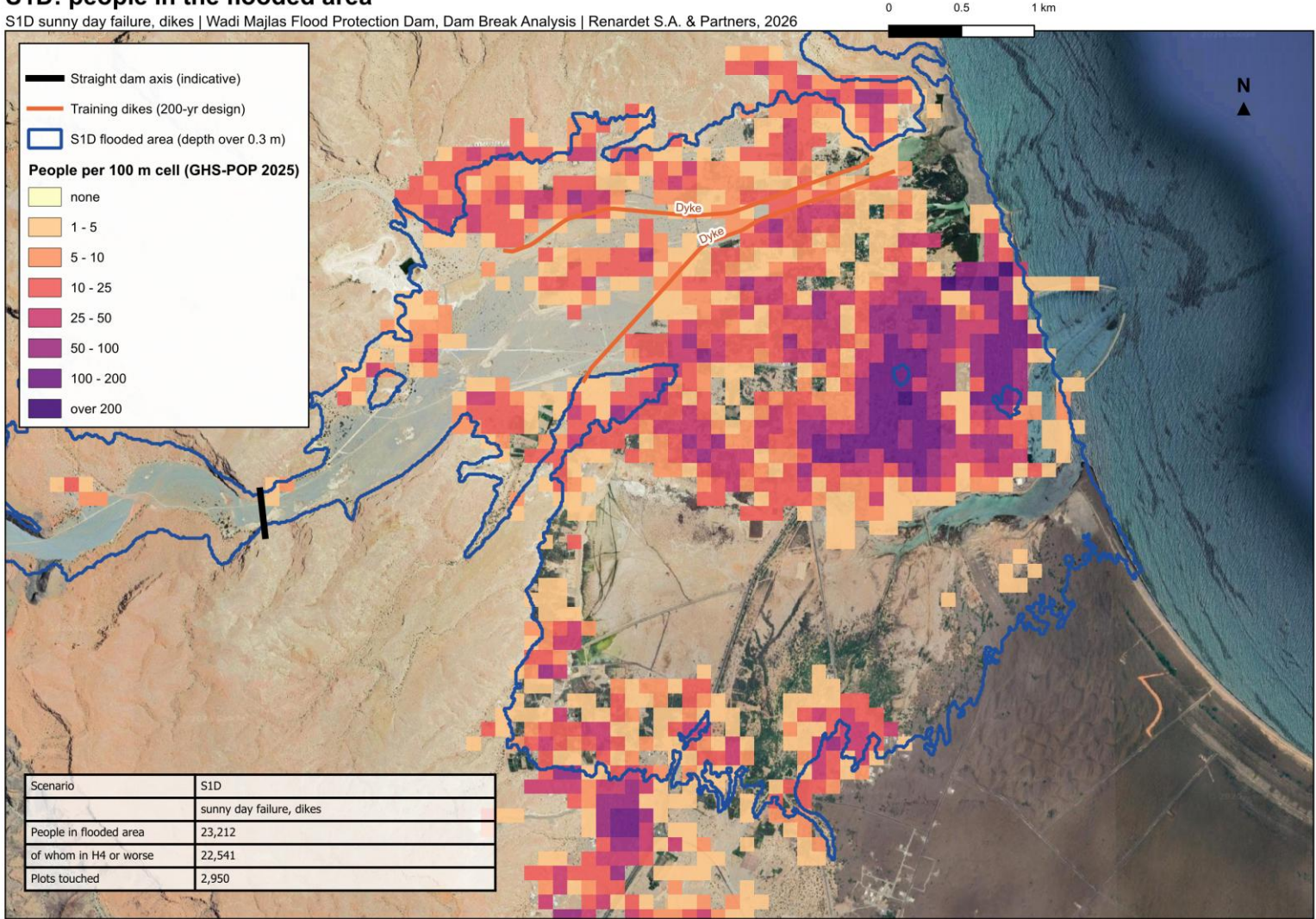


Figure 31 - S1D: people per 100 m cell (GHS-POP 2025) with the flooded area outlined.

S2D: people in the flooded area

S2D flood day PMF failure, dikes | Wadi Majlas Flood Protection Dam, Dam Break Analysis | Renardet S.A. & Partners, 2026

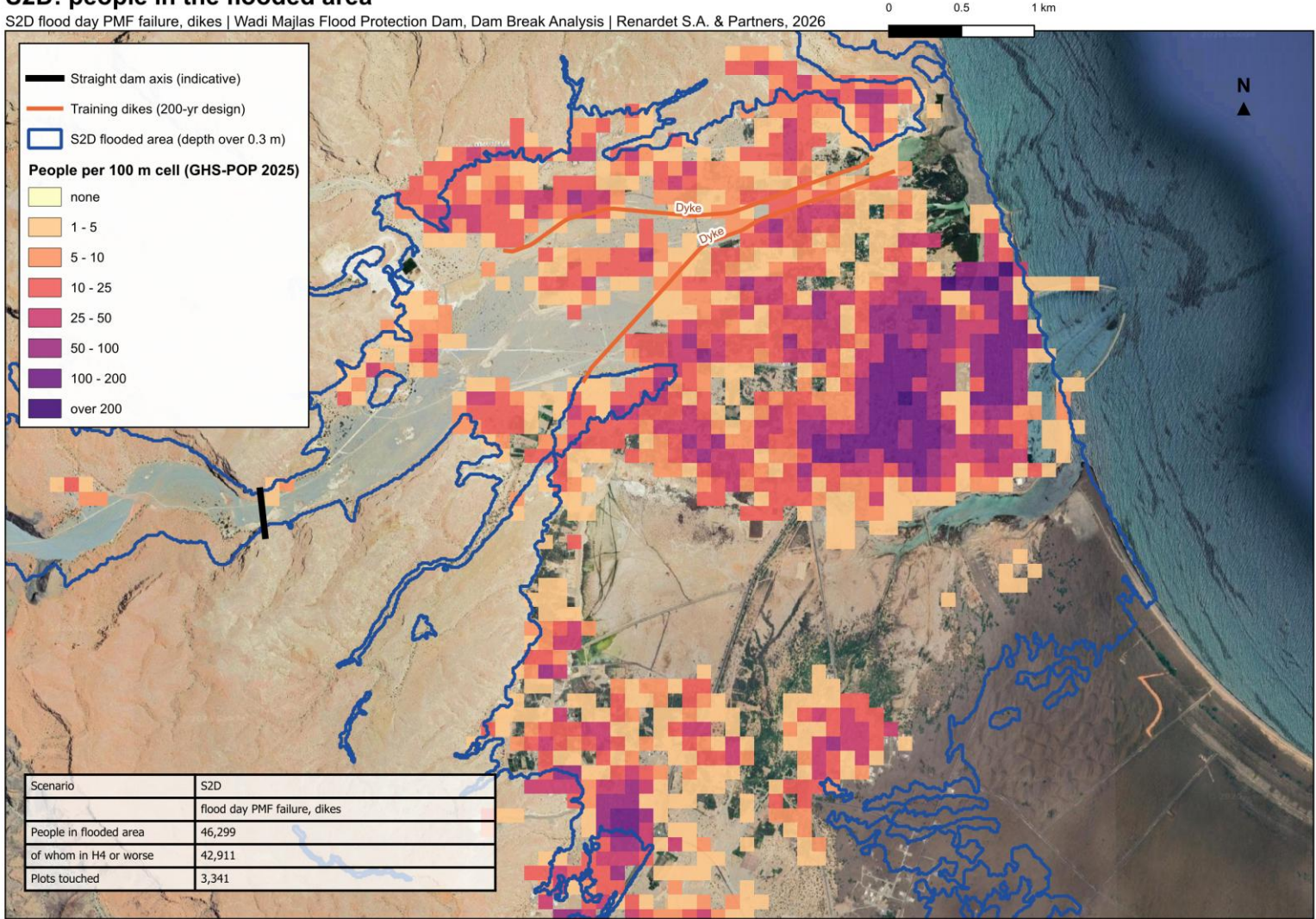


Figure 32 - S2D: people per 100 m cell (GHS-POP 2025) with the flooded area outlined.

S3D: people in the flooded area

S3D PMF no failure, dikes | Wadi Majlas Flood Protection Dam, Dam Break Analysis | Renardet S.A. & Partners, 2026

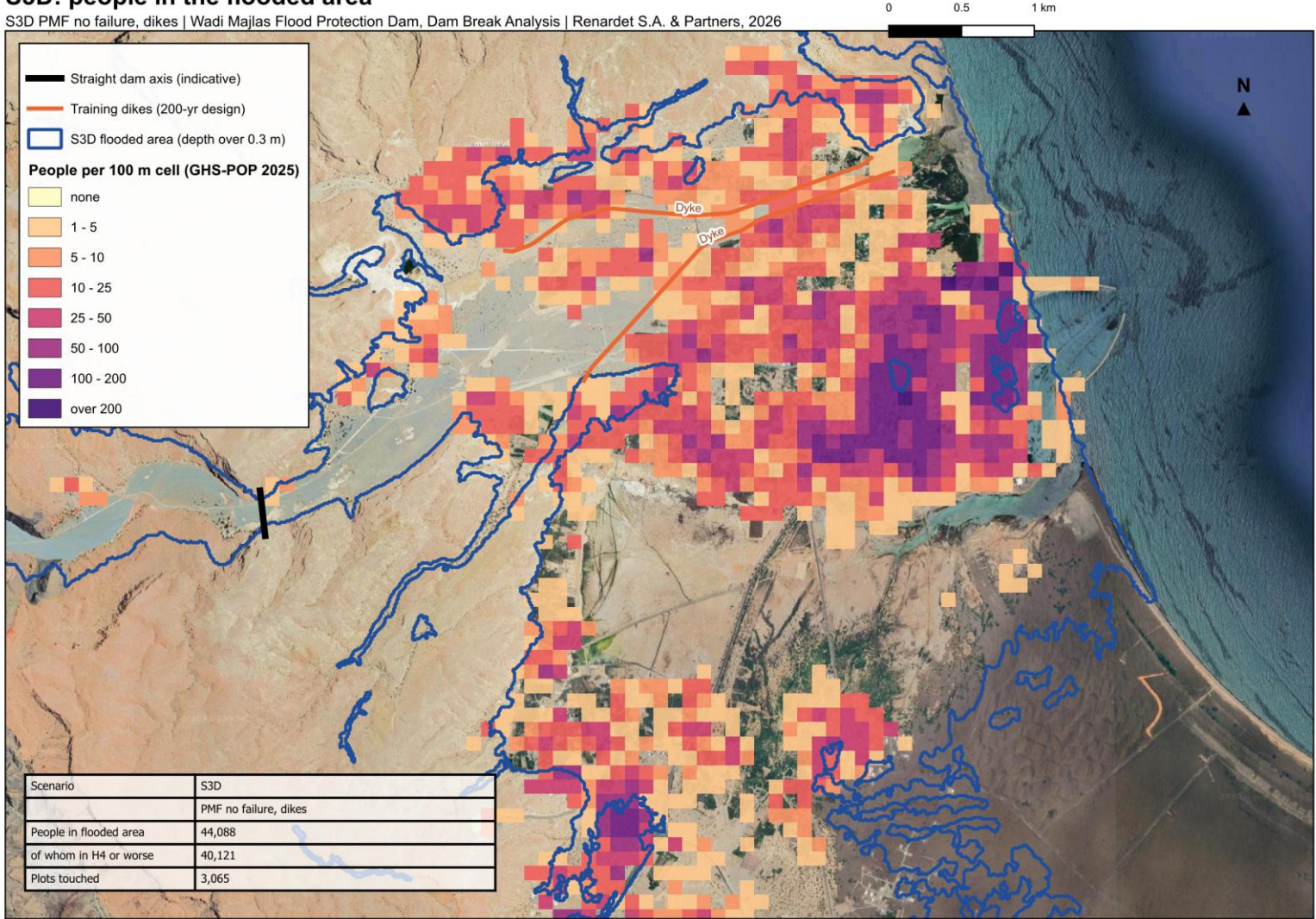


Figure 33 - S3D: people per 100 m cell (GHS-POP 2025) with the flooded area outlined.

S2D: arrival of the failure wave over the PMF (0.3 m extra depth)

S2D flood day PMF failure, dikes | Wadi Majlas Flood Protection Dam, Dam Break Analysis | Renardet S.A. & Partners, 2026

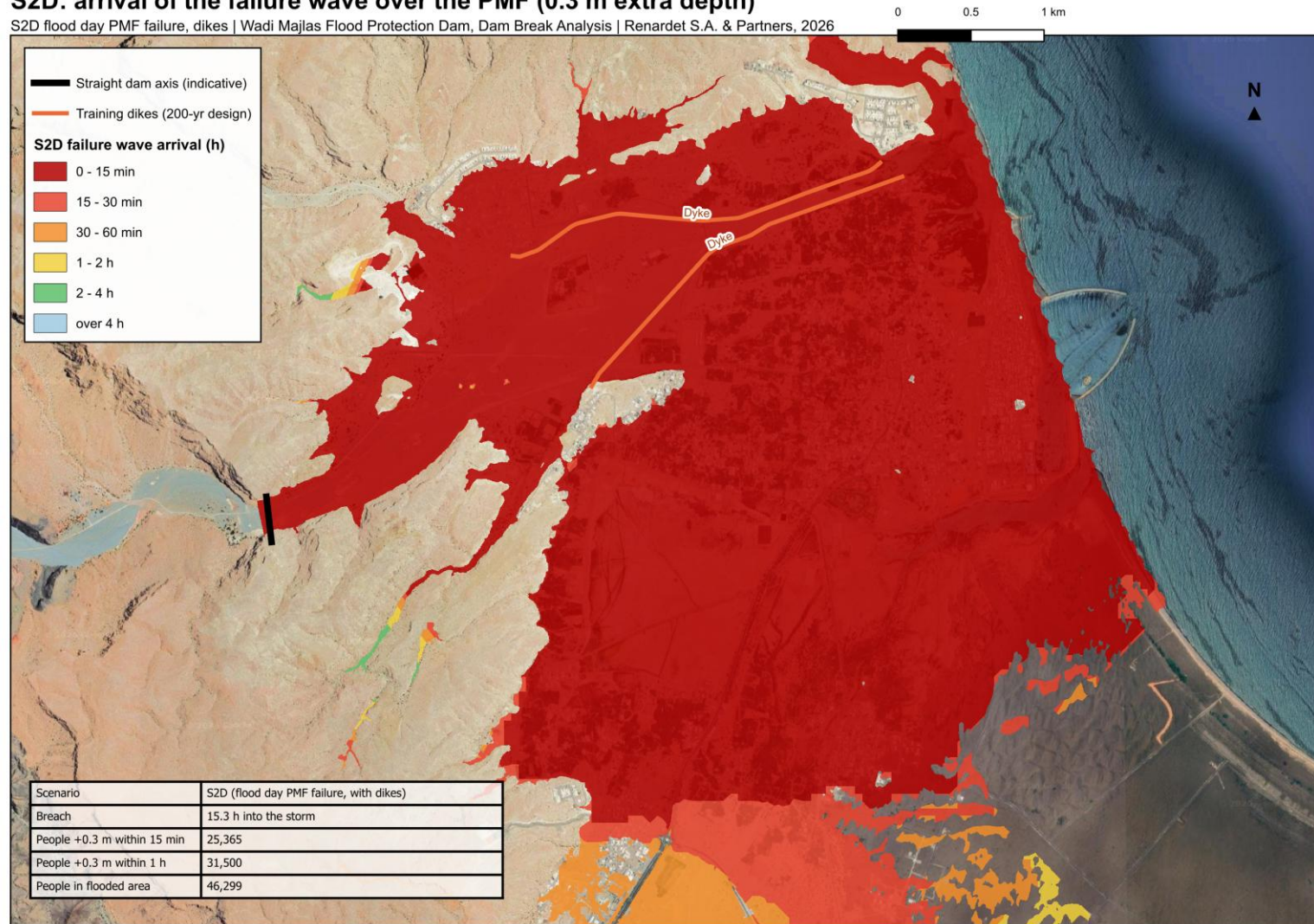


Figure 34 - S2D: arrival of the failure wave over the PMF, the time after the breach at which the water is 0.3 m deeper than without failure.

10. DISCUSSION

10.1. WHAT THE FAILURE ADDS TO THE FLOOD

The flood-day runs are read against the no-failure twins: the difference in depth, arrival time and hazard class is the effect of the dam, and only that difference belongs to the dam-safety classification. The sunny-day runs stand alone: there is no flood to compare with, and everything in the flooded area is the dam's doing.

10.2. THE TRAINING DIKES

The dikes were designed to pass the 200-year wadi flood. A dam break wave is one to two orders of magnitude larger; the maps show it overtopping them within minutes. The dike runs and the no-dike runs are both reported so that the design review can judge whether the dikes change where the water goes, not whether they contain it.

10.3. LIMITS OF THE ANALYSIS

- The breach is prescribed, not computed: its width and speed are assumptions bounded by the sensitivity runs.
- The terrain of the plain is the 5 m drone survey; buildings are not resolved, so depths in the town are those of an open plain with the town's roughness.
- The population grid is a 2025 estimate at 100 m; the plot layer is the flood-risk study layer.
- The sea level is fixed at mean higher high water, 0.71 m a.s.l.; a storm surge coinciding with the PMF was not modelled.



11. CONCLUSIONS AND RECOMMENDATIONS

- A sunny-day failure of five monoliths releases 103 Mm^3 with a peak of $45,312 \text{ m}^3/\text{s}$; the coastal plain is reached within 15 min and 23,212 people live in the flooded area. Warning must therefore come from the dam, before the failure, not from the flood.
- The PMF without failure peaks at 96.06 m a.s.l. with $17,323 \text{ m}^3/\text{s}$ over the spillway; the dam is not overtopped.
- A failure at the peak of the PMF raises the peak flow from 17,323 to $63,858 \text{ m}^3/\text{s}$ and the flooded area only from 25.9 to 27.1 km^2 ; the plain is already under the PMF, so the dam adds depth and force rather than extent (people in the flooded area 44,088 against 46,299).
- Time, not extent, separates the scenarios: the sunny-day wave reaches 15,579 people within 15 minutes of the breach and 23,179 within an hour, while the PMF without failure takes hours to build up. Warning must therefore come from the dam's own monitoring, and the plan should use the arrival bands of chapter 9 as its warning zones.
- The emergency action plan should use the hazard maps for evacuation routes and shelters.
- The training dikes should not be relied on in the plan: they are overtopped by every failure scenario.
- The breach width is the parameter that matters for the peak; the formation time matters for the first quarter hour of warning; the sensitivity runs bound the results and should be quoted with them.



REFERENCES

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- US Army Corps of Engineers (2024). HEC-RAS 6.6 User's Manual and 2D Modeling User's Manual. Hydrologic Engineering Center, Davis.
- FERC (2014). Engineering Guidelines for the Evaluation of Hydropower Projects, Chapter 2, Selecting and Accommodating Inflow Design Floods for Dams.
- USACE (2014). Using HEC-RAS for Dam Break Studies, TD-39. Hydrologic Engineering Center.
- AIDR (2017). Guideline 7-3: Flood Hazard. Australian Institute for Disaster Resilience.
- ICOLD (2020). Bulletin 111, Dam Break Flood Analysis, Review and Recommendations.



APPENDIX A: RUN REGISTER

One line per HEC-RAS plan: the scenario it computes, the solver's own completion status, the wall-clock time and the largest Courant number reached. The base runs had the 32-thread workstation to themselves; from the sensitivity phase on, three runs shared it, so their times are not comparable.

Table 19 - Run register.

Scenario	HEC-RAS plan	Solver status	Run time	Max Courant
Fill	p27	restart file written	-	-
FillN	p43	restart file written	-	-
S3D	p32	Unsteady Finished Successfully	05:23:35	0.88
S3N	p33	not run: dikes immaterial (chapter 3)	-	-
S1D	p28	Unsteady Finished Successfully	01:06:11	0.83
S1N	p29	Unsteady Finished Successfully	01:46:55	0.80
S2D	p30	Unsteady Finished Successfully	09:24:00	0.88
S2N	p31	not run: dikes immaterial (chapter 3)	-	-
S1D-W51	p37	Unsteady Finished Successfully	01:56:42	0.80
S1D-W153	p38	Unsteady Finished Successfully	01:08:22	0.89
S1D-C1.44	p41	Unsteady Finished Successfully	00:49:40	0.80
S1D-T3m	p44	Unsteady Finished Successfully	02:06:58	0.80
S1D-T18m	p45	Unsteady Finished Successfully	02:04:19	0.80
S2D-W119	p35	Unsteady Finished Successfully	06:45:00	0.88
S3D-F10000	p48	Unsteady Finished Successfully	14:58:00	0.89
S2D-F10000	p42	pending	-	-
S2D-W51	p34	not run: width tested on the sunny day and once on the flood day (S2D-W119)	-	-
S2D-W153	p36	not run: width tested on the sunny day and once on the flood day (S2D-W119)	-	-
S2D-T18m	p39	stopped at 36 %: formation time tested on the sunny day (S1D-T18m)	-	-
S2D-T3m	p40	stopped at 24 %: formation time tested on the sunny day (S1D-T3m)	-	-



APPENDIX B: MODEL FILES

Where the model, the results and the figures are kept, for whoever picks the study up next.

- HEC-RAS project 'Majlas' in the study folder: geometries 'Dam Breack' (dikes) and 'Dan Breack - without dikess' (no dikes); flow files u26 to u31; plans p27 to p43 and p48, named after the scenarios.
- Results per scenario in W2/04_data/results/<plan>/: summary.json, connection.csv (flow, pool, breach), boundary series, rasters of maximum depth, velocity, depth times velocity, arrival time, duration and AIDR hazard class (2 m, EPSG:32640).
- Maps in W2/02_figures/maps/results/, charts in W2/02_figures/charts/, the QGIS project W2/03_maps/Majlas_DamBreak_W2.qgz with the layouts.

